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A Millimeter-Wave CMOS Wideband Concurrent Dual-Band Dual-Beam Phased array Receiver Front-End

Kamran Entesari

Outline

I. Introduction
II. System Architecture
  A. Array literature review
III. Proposed Array Architecture
  A. Circuit Implementation
    1. Low Noise Amplifier (LNA)
    2. Phase Shifter (PS)
    3. IQ to VMs Interface
    4. Interface of VM to Combiner
    5. Wideband Distributed Active Combiner
  B. Fabrication and Measurement
IV. Conclusion

Outline

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Introduction

- 5th Generation wireless communication:
  - Faster
  - More reliable connectivity
  - Supporting more devices
  - Supporting new cases: Smart Cities, Autonomous Vehicles, Entertainment, Agriculture

Introduction

- 5G wireless communication systems (Realizing)
- Multi-antenna approach
  - Compensating free path loss
  - Spatial selectivity -> Enabling multiple data streams
  - Spatial interfere cancelation (spectrum interfere cancelation??)
- MIMO (Multiple-Input Multiple-Output).
  - Single-user MIMO for high data rate.
  - Multi-user MIMO for more user capacity.

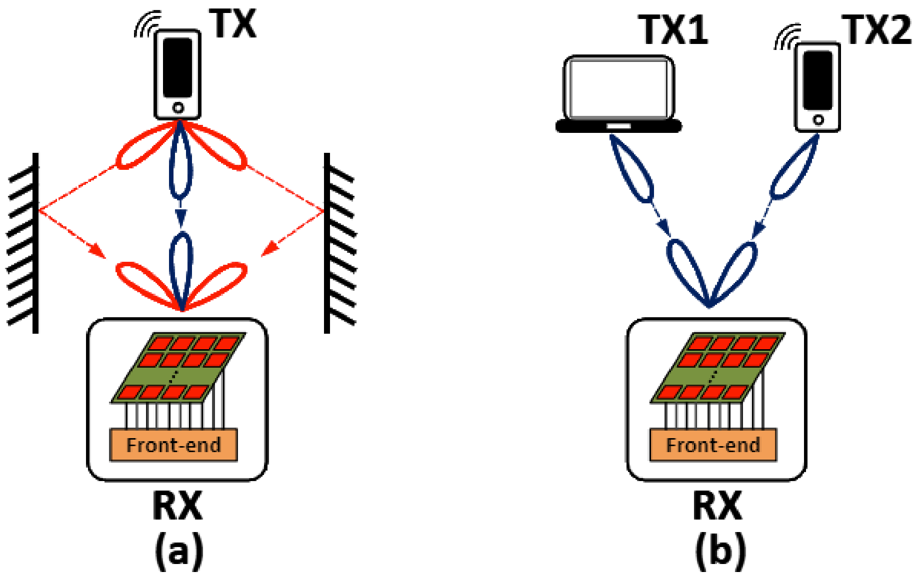

Fig. 1. Multi-antenna different applications in MIMO, (a) SU-MIMO. (b) MU-MIMO.

Introduction

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- 5G wireless communication systems (Frequency)
- Sub-6G
  - 410 – 7100 MHz, including multiple sub-bands.
- mm-wave bands
  - N257: 26.5 – 29.5 GHz
  - N258: 24.25 – 27.5 GHz
  - N259: 39.5 – 43.5 GHz
  - N260: 37 – 40 GHz
  - N261: 27.5 – 28.35 GHz

Introduction

- MIMO for 5G: beamforming: 1) wideband/reconfigurable/multi-frequency, 2) multi-stream

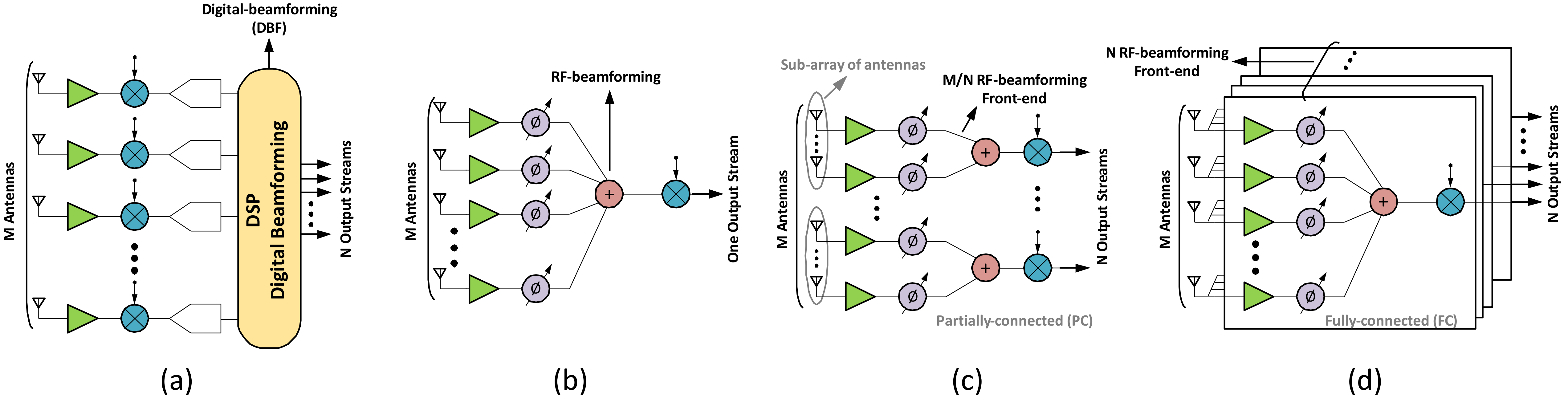

Fig. 2. Different types of beamforming, (a) Digital-beamforming (DBF) supporting N output streams. (b) RF-beamforming (RFB) supporting only one output stream. (c) RF-beamforming supporting N output streams.

Introduction

- To achieve **multi-beam**, **wideband/reconfigurable/multi-band** MIMO, A mm-Wave Concurrent Dual-Band Dual-Beam Phased array Receiver is proposed:
  - Major contribution in both system-level and circuit-level
  - Fully integrated mm-Wave concurrent dual-band dual-beam phased array RX front-end

Outline

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Literature review (Array)

- 16-Element fully integrated 28-GHz Digital beamforming receiver.
- 4 simultaneous beam.
- Mixer, and ADC should be highly linear.

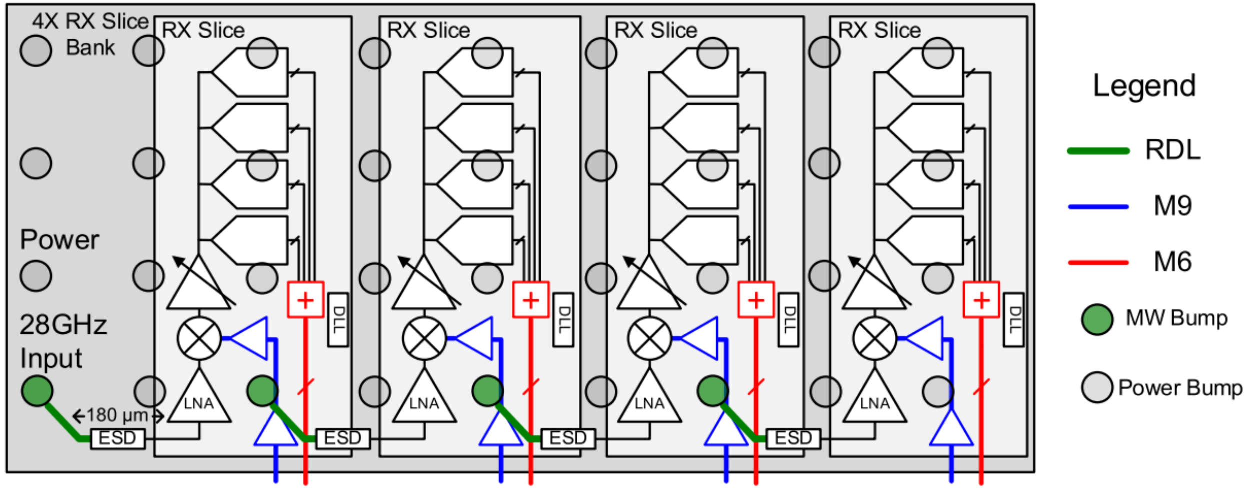

Fig. 3. Placement and signal routing of a 4 × RX slice bank.

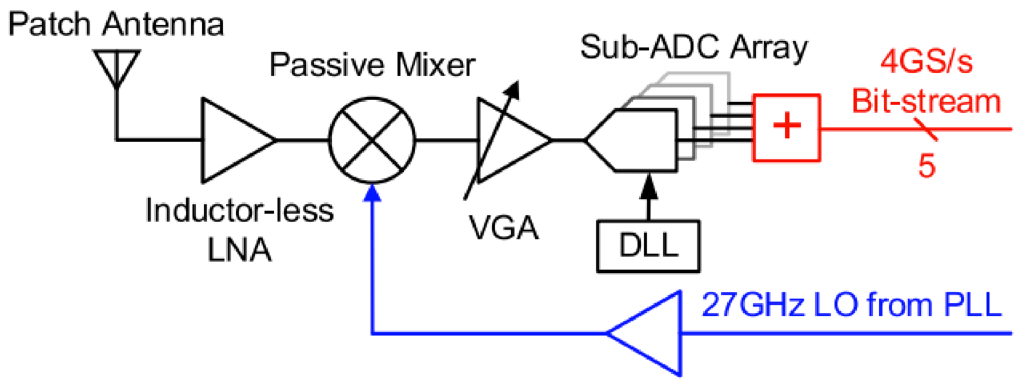

Fig. 4. Block diagram of a single RX slice.

R. Lu, C. Weston, D. Weyer, F. Buhler, D. Lambalot, and M. P. Flynn, “A 16-element fully integrated 28-GHz digital RX beamforming receiver,” IEEE Journal of Solid-State Circuits, vol. 56, no. 5, pp. 1374–1386, 2021

Literature review (Array)

- A 25-30 GHz FC hybrid-beamforming receiver.
- 2 simultaneous received output signal streams.
- Both in RF-domain and DSP in digital-domain.
- Cartesian phase-shifting techniques done in down-conversion.
- No need any quadrature network, inherently wideband.
- Doubling the number of mixers that are needed in down-conversion

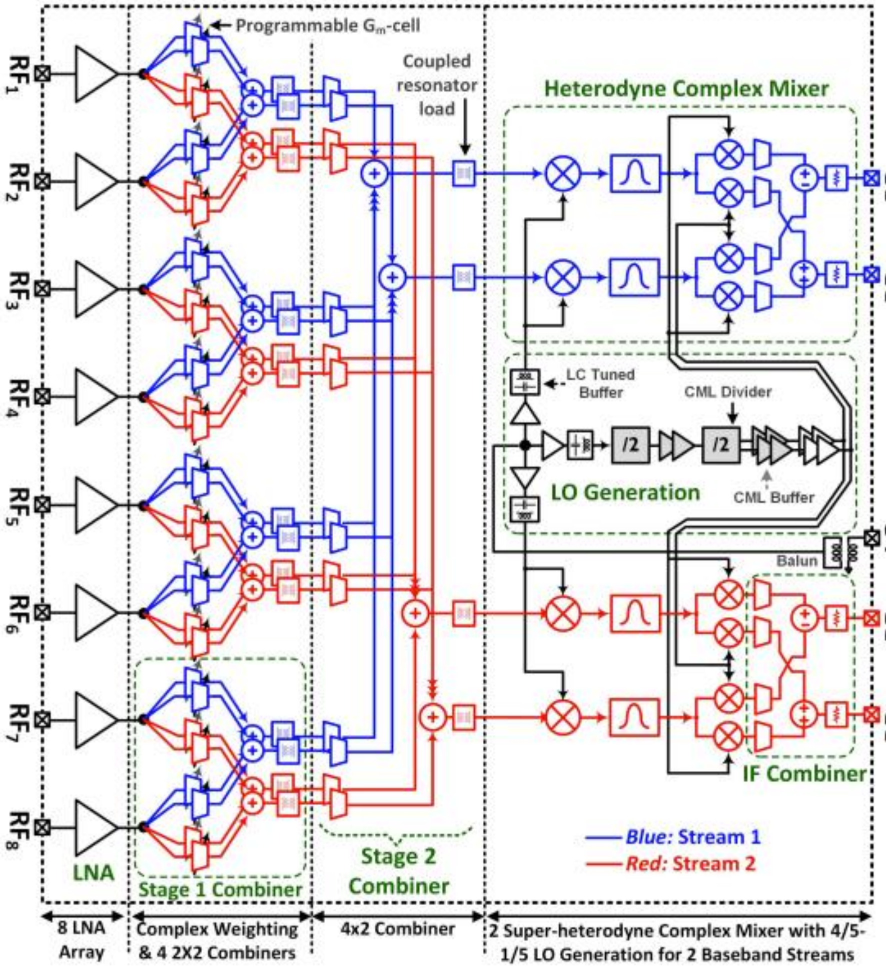

Fig. 4. Eight-element two-stream wideband heterodyne Cartesian-combining hybrid beamforming prototype.

S. Mondal, R. Singh, A. I. Hussein, and J. Paramesh, “A 25–30 GHz fully-connected hybrid beamforming receiver for MIMO communication,” IEEE Journal of Solid-State Circuits, vol. 53, no. 5, pp. 1275–1287, 2018

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Proposed Array Architecture

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- 4 input channels and 2 output streams
- Fully-connected RF beamforming
- Dual-band
  - ➢ 24 – 28GHz
  - ➢ 37 – 40GHz
  - ➢ Carrier aggregation mode (CA)
- Dual-beam
- Concurrent
- Null-tuning

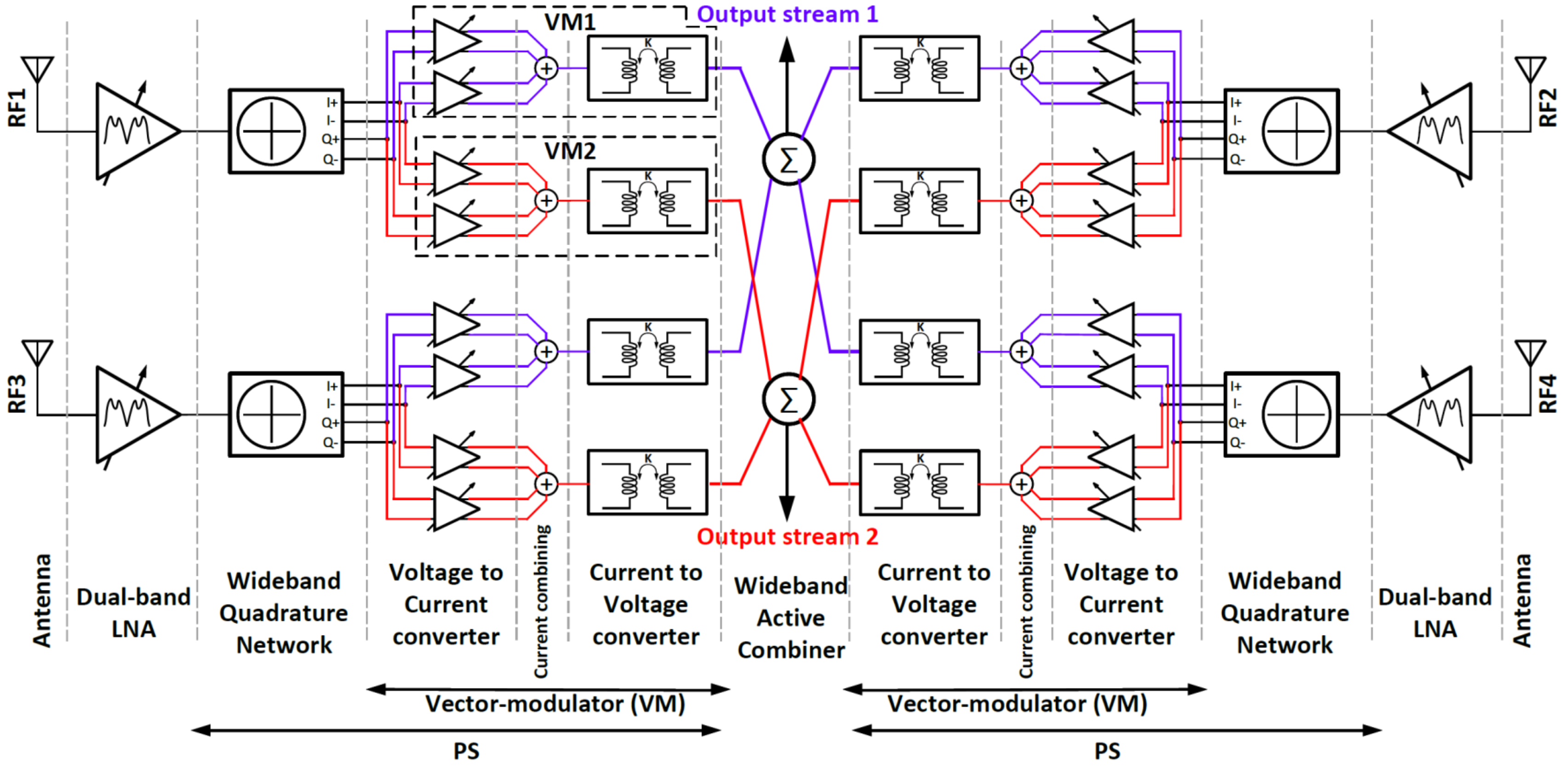

Fig. 14. Block diagram of the proposed mm-Wave concurrent dual-band dual-beam phased-array receiver front-end.

Proposed Array Architecture

- Functionality

1. Both output streams operating at 28 GHz
2. Both output streams operating at 39 GHz
3. Carrier-aggregation (CA) mode; One output stream operates at 28 GHz and the other operates at 39 GHz

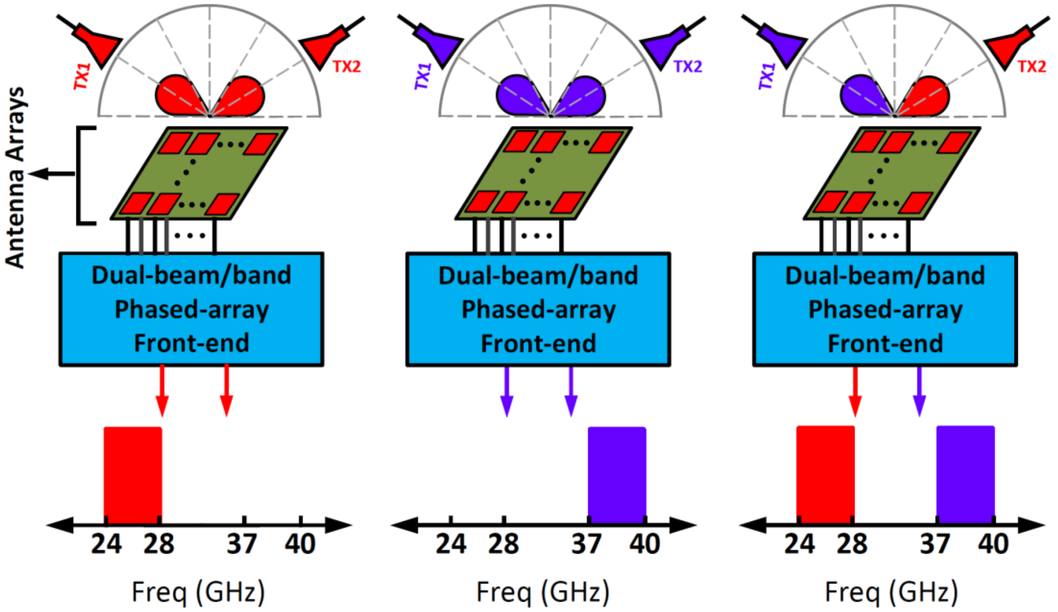

Fig. 15. The concurrent dual-band dual-beam phased-array receiver different functionalities.

Proposed Array Architecture

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- System level simulations

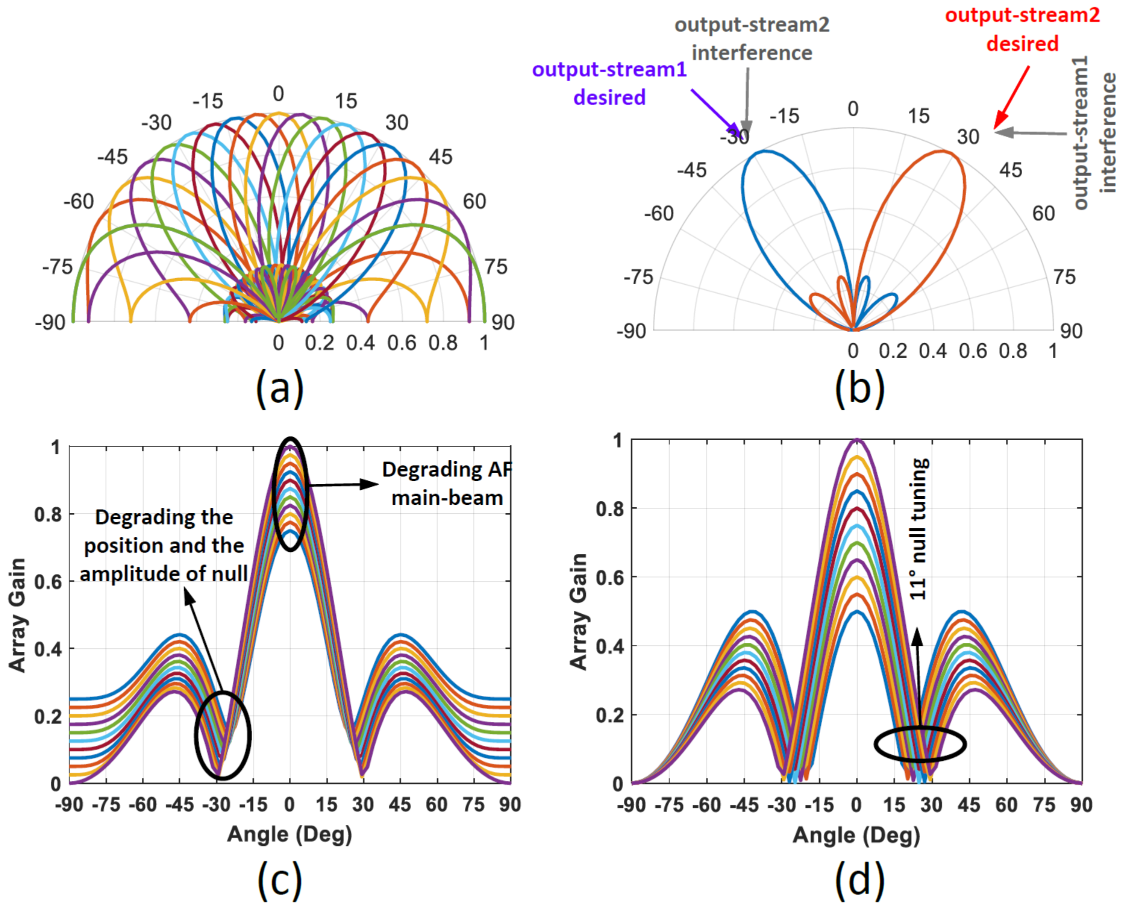

Fig. 16. System level simulations at 28 GHz. (a) single beam steering with 11.25◦ steps a 28 GHz. (b) concurrent dual-beam functionality. (c) The effect of gain mismatch on the AF. (d) RF null-tuning with main-beam at 0◦.

Proposed Array Architecture

- Link Budget

$$P_{RX} = P_{TX} + G_{TX,ANT} + 10\,log(N^2_{TX,ANT}) - L_{FS} + G_{RX,ANT} + 10\,log(N^2_{RX,ANT})$$

$$P_{RX,min} = -174\,\frac{dBm}{Hz} + 10\,log(BW_c) + SNR_{out} + NF_{RX} + Link\,Margin$$

$$(SNR)_{out} = 20\,log(\frac{1}{EVM})$$

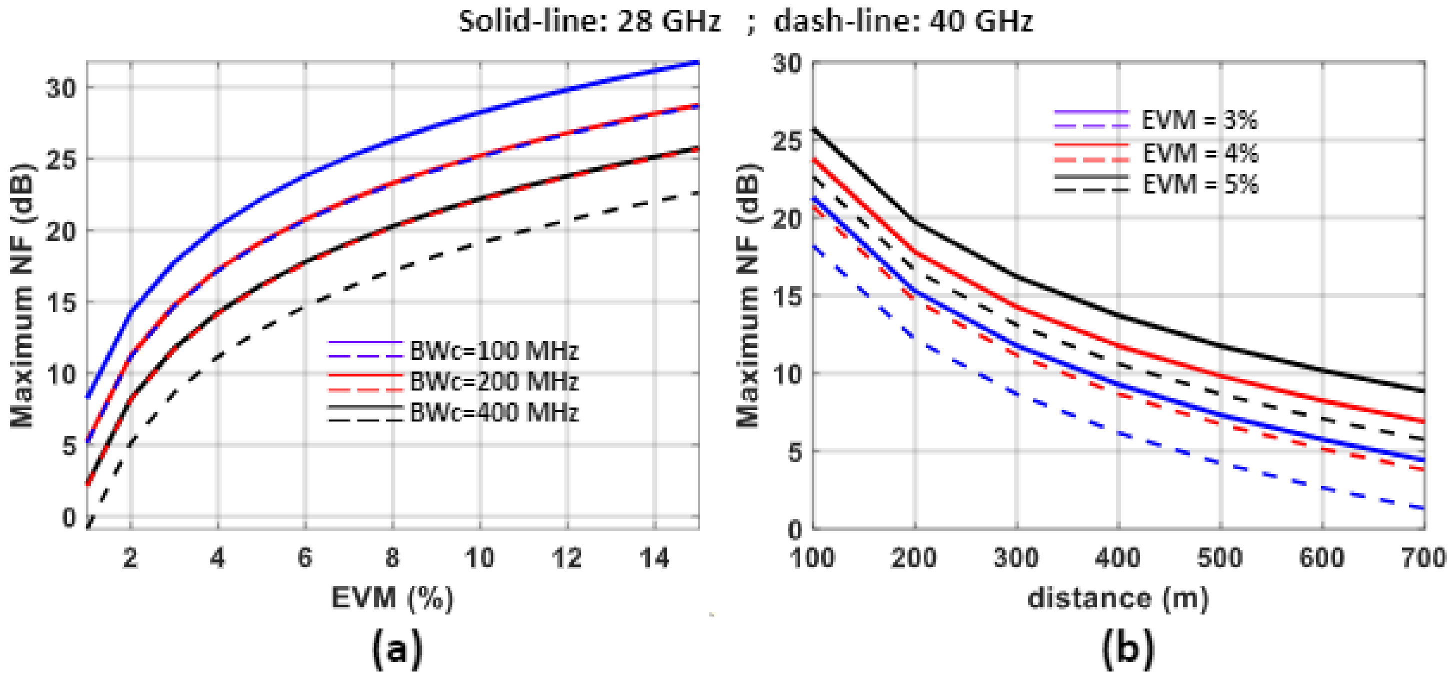

Fig. 17. (a) The required NF for the receiver per channel versus different EVM percentage for different channel bandwidth at 300m distance for communication. (b) The required NF for the receiver versus distance for different EVM percentage at 400 MHz channel bandwidth.

Proposed Array Architecture

- The effect of Number of Bits of PS on EVM

$$Max - EVM - degradation = N \times [\frac{\sin(\frac{2*\pi}{4\times 2^{N_{ps}}})}{\sin(N \times \frac{2*\pi}{4\times 2^{N_{ps}}})}]$$

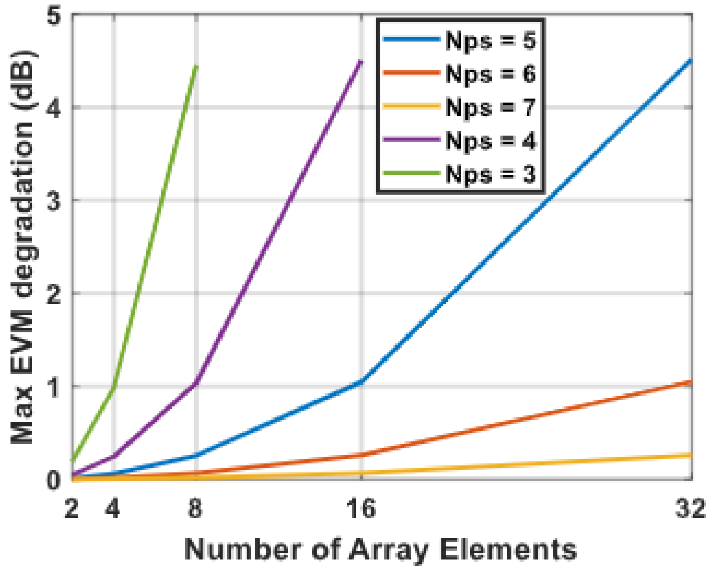

Fig. 18. The effect of phase step error on EVM

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Circuit Implementation

- 4 Low noise amplifiers (LNA)
  - The majority of this block is done by Jierui Fu.
- 4 Phase shifters (PS)
  - 4 Quadrature network (QN)
  - 8 Vector-modulator (VM)
- 2 Combiners
- Global Foundry 22 nm Fully Depleted Silicon On Insulator (FDSOI) CMOS

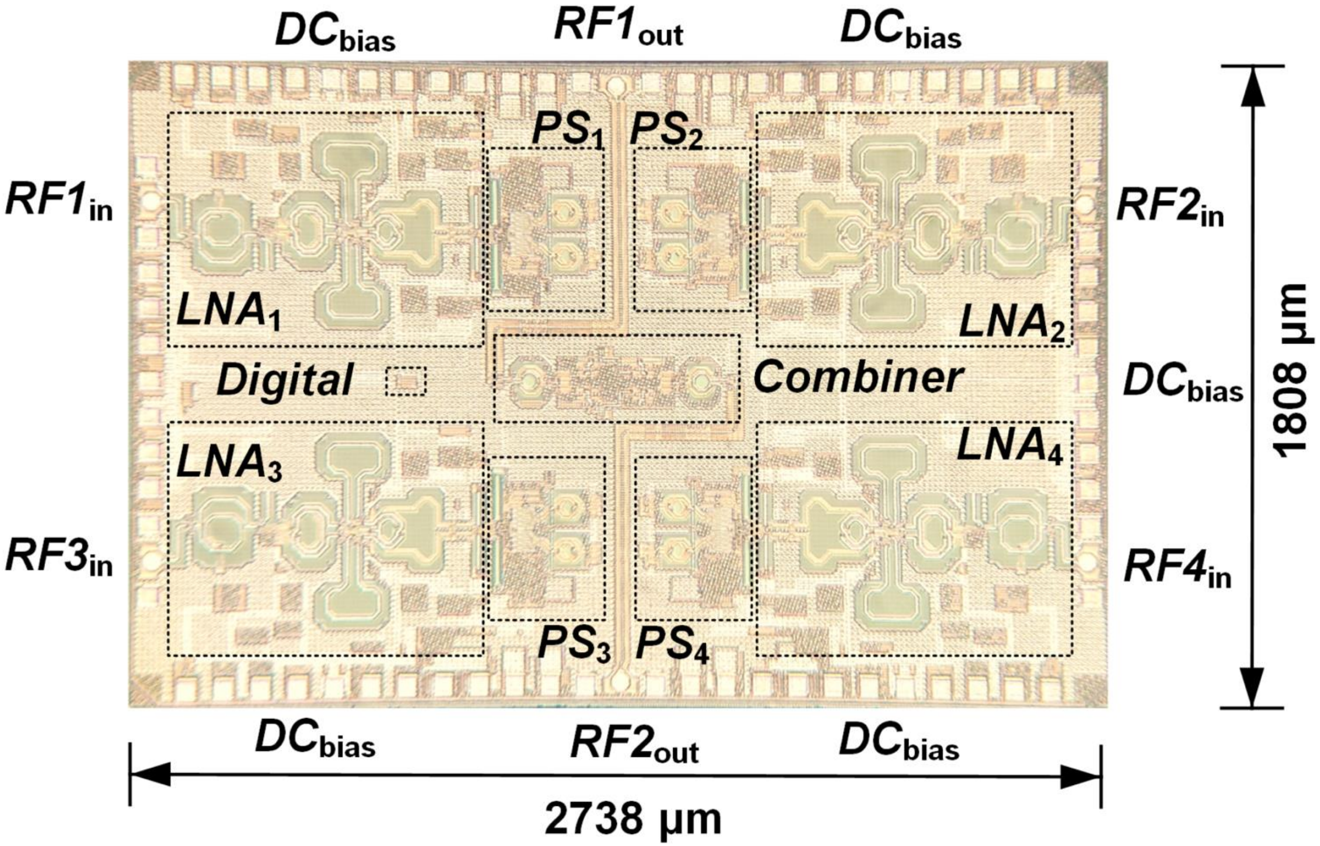

Fig. 19. Die photo of the concurrent dual-band dual-beam phased-array receiver.

Low Noise Amplifier (LNA)

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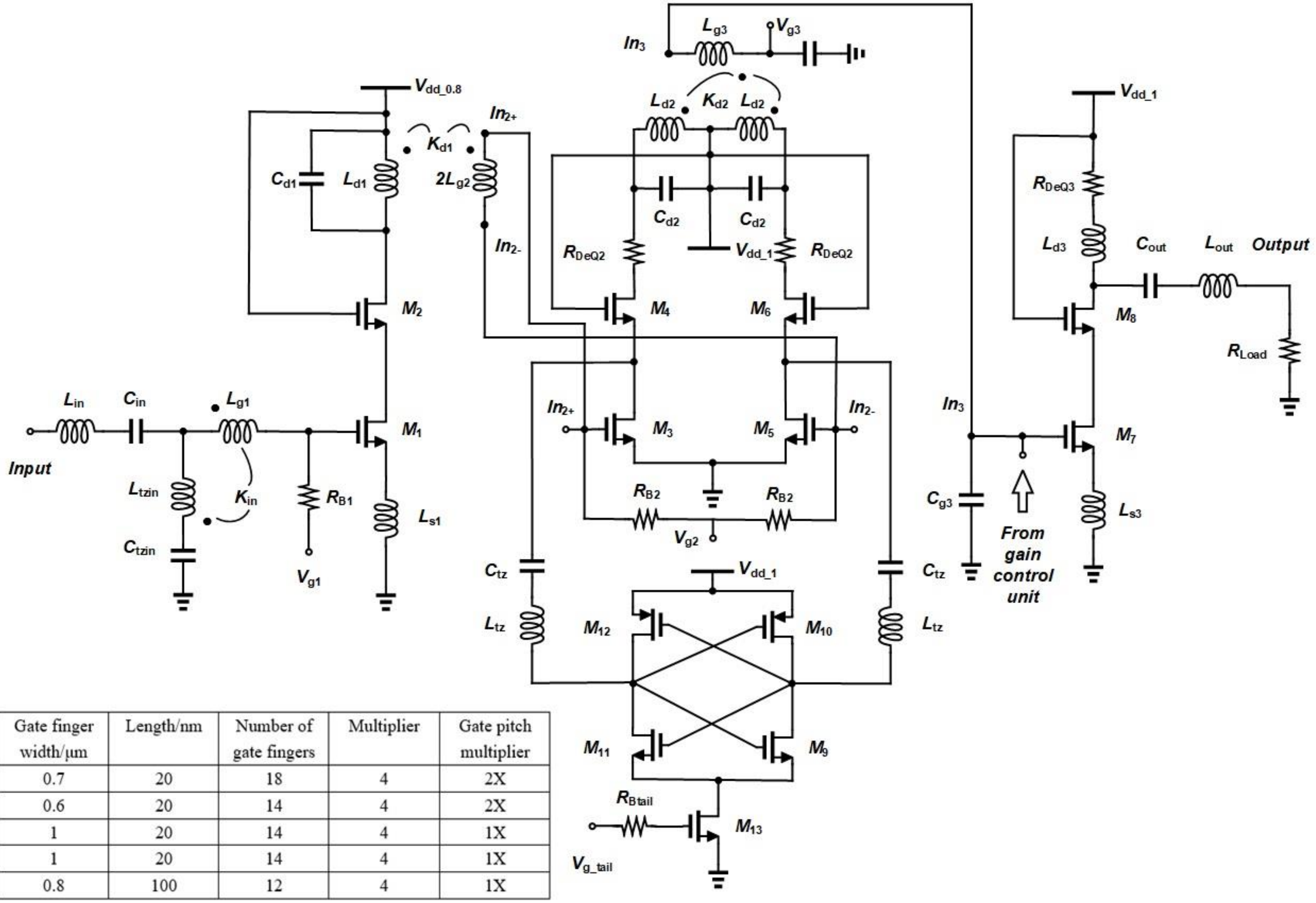


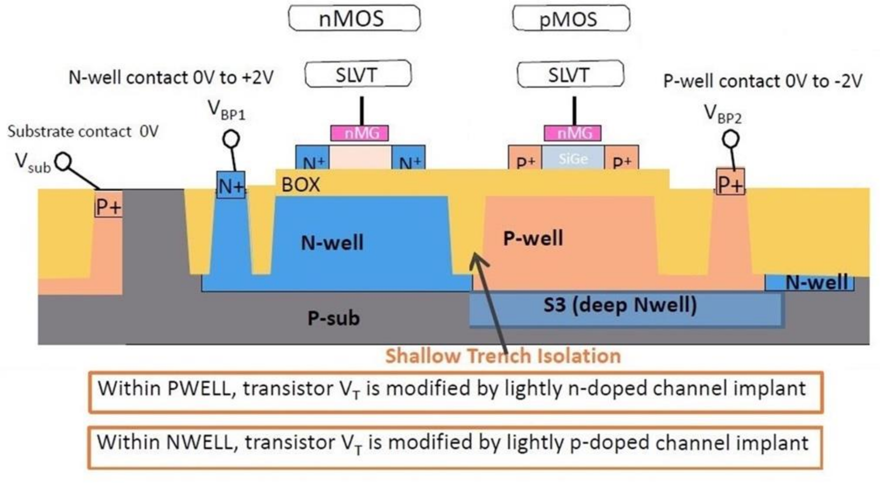
Transistor	Gate finger width/μm	Length/nm	Number of gate fingers	Multiplier	Gate pitch multiplier
M1/M2	0.7	20	18	4	2X
M3-M8	0.6	20	14	4	2X
M9/M11	1	20	14	4	1X
M10/M12	1	20	14	4	1X
M13	0.8	100	12	4	1X

Component	Value
L_{in}	69 pH
L_{tzin}	360 pH
L_{g1}	180 pH
L_{s1}	55 pH
L_{d1}	220 pH
L_{g2}	380 pH
L_{d2}	155 pH
L_{tz}	335 pH
L_{g3}	210 pH
L_{s3}	20 pH
L_{d3}	160 pH
L_{out}	47 pH
C_{in}	526 fF
C_{tzin}	419 fF
C_{d1}	136 fF
C_{d2}	98 fF
C_{tz}	72 fF
C_{g3}	39 fF
C_{out}	269 fF
k_{in}	0.065
k_{d1}	0.58
k_{d2}	0.2
R_{DeQ2}	19 Ω
R_{DeQ3}	35 Ω
R_{B1}	3 kΩ
R_{B2}	1.7 kΩ
R_{Btail}	4.1 kΩ

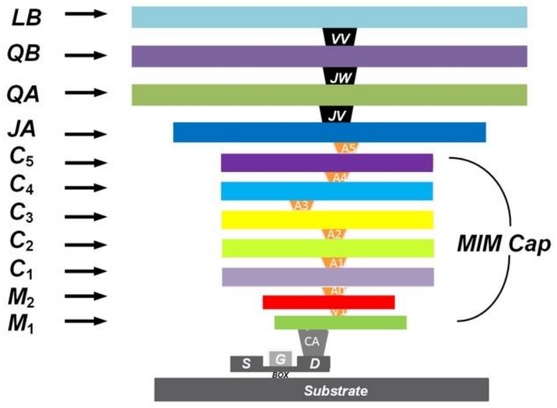
Fig. 21. Schematic of the concurrent LNA.

III. A. LNA Design 6) Active devices

- 22 nm FDSOI CMOS process from Global Foundries.
- The Super low Vt (SLVT) transistor
  - Lower noise and better high-frequency performance.
- Forward body biasing (FBB)
  - A Positive voltage on the body of NMOS, VBP1.
  - A negative voltage on the body of PMOS, VPB2.
- The metal stack option 19
  - 11 layers of metals
  - The topmost aluminum metal (LB)



(a)



(b)

a) The cross section of FDSOI flip-well SLVT transistors, and (b) the metal stacks of 22nm FDSOI Option 19.

III. A. LNA Design 6) Active devices

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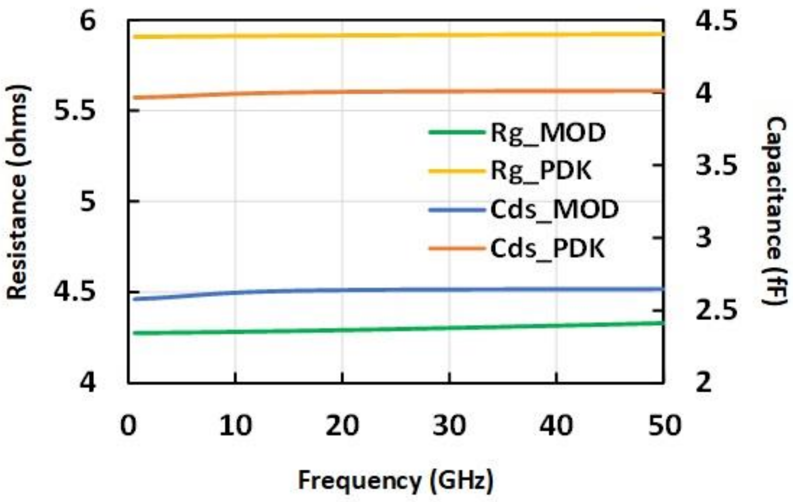
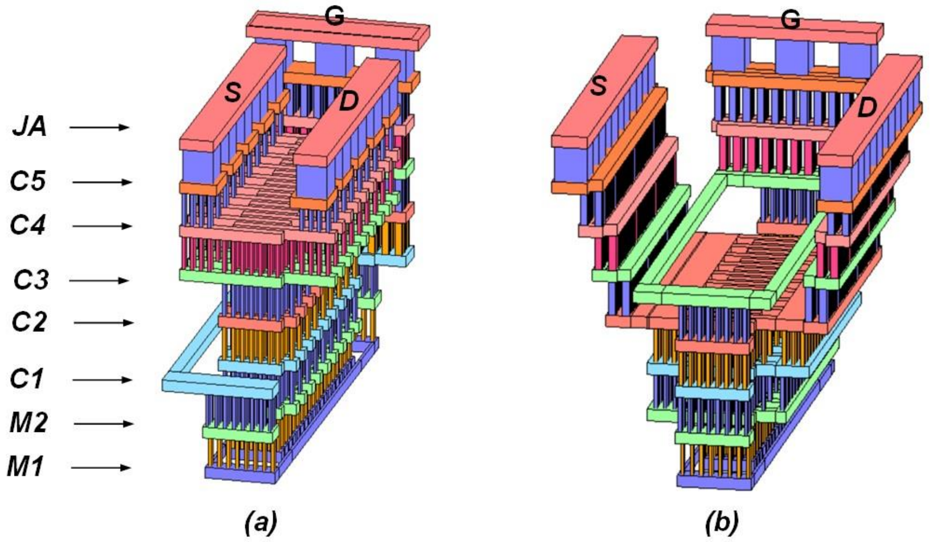
- The standcell of first stage input NMOS transistor
  - A relaxed 2X pitch to reduce the parasitic capacitance.
  - A multiple of 4, as in 2 by 2, to reduce the routing resistance and improve matching during fabrications.
  - A multiple finger number to reduce the gate resistance.
- The layout is modified to a staircase pattern



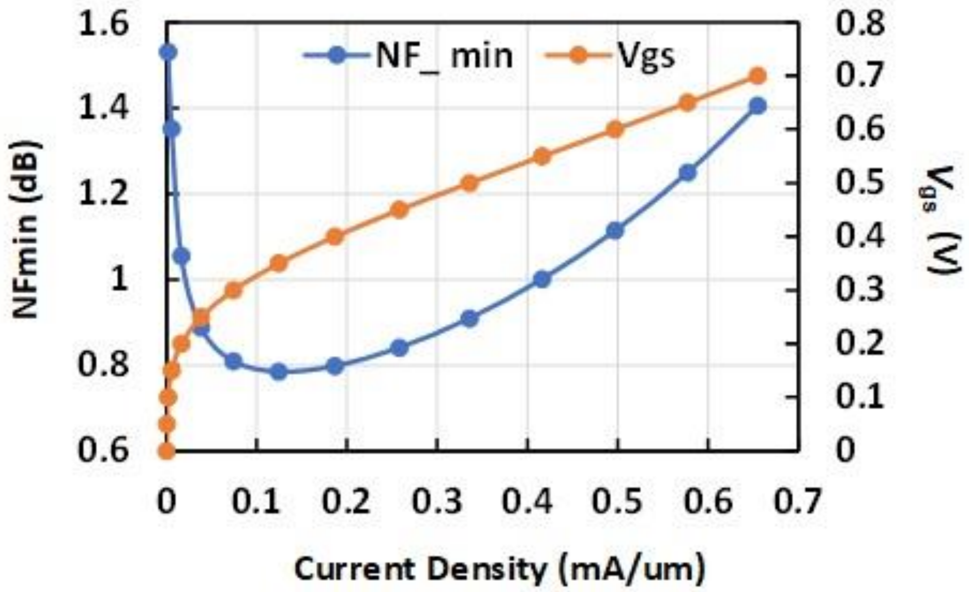
Cds and Rg due to metal routing.



(a) The layout of standcell for one of the multiplier (out of four) with finger width WF = 0.7μm and number of finger NF = 18 from PDK, (b) The layout of standcell with modification.

III. A. LNA Design 6) Active devices

- For a current density of 0.1 to 0.2 mA/µm, the NF reaches its lowest value.



Simulated NFmin and Vgs with respect to Current Density.

Low Noise Amplifier (LNA)

- 3-stage cascode design.
- Single-ended input and output stage.
- An active notch in the differential 2nd stage.
- 8-step 7-dB gain control in the 3rd stage.
- 0.8 V, 1.0 V, and 1.0 V VDD for each stage.

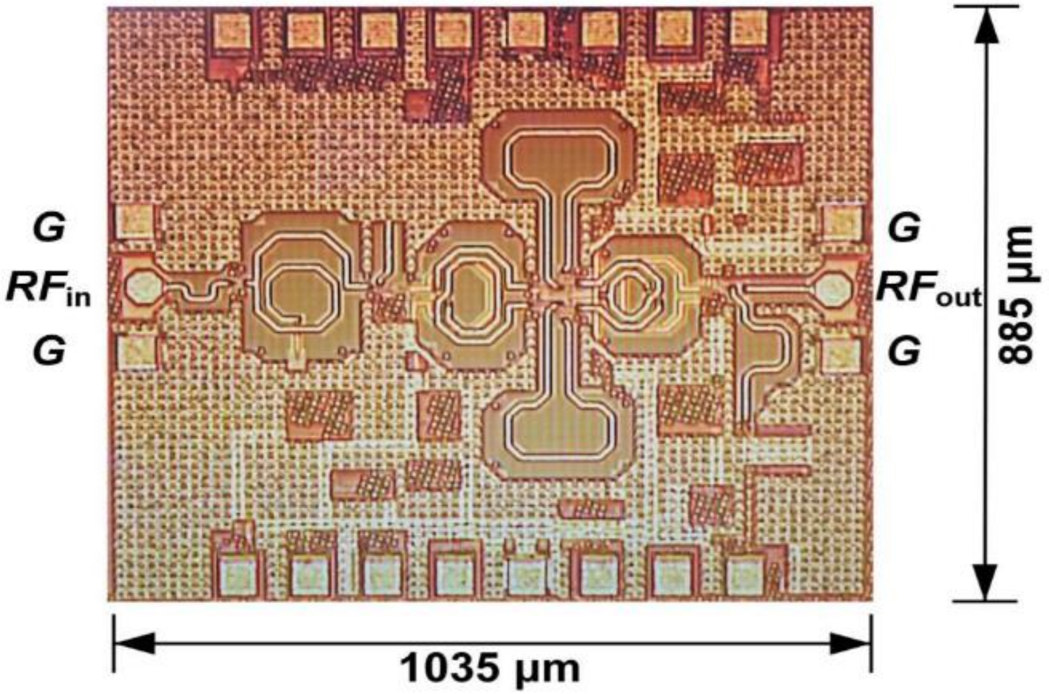
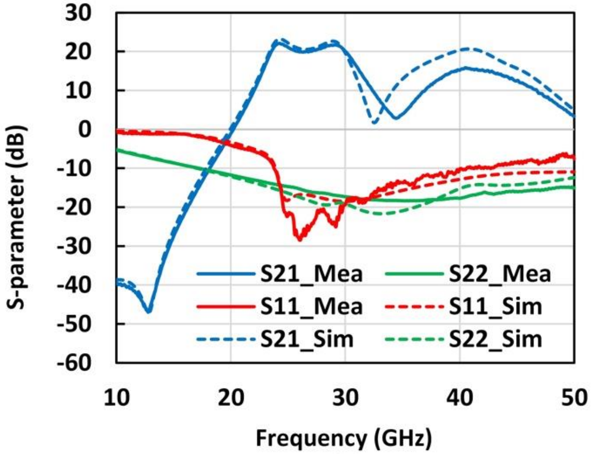


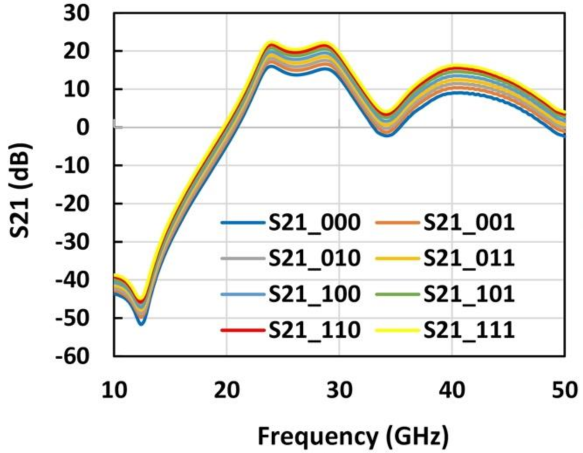
Fig. 20. Die photo of the concurrent dual-band stand-alone LNA

IV. Fabrication and measurement

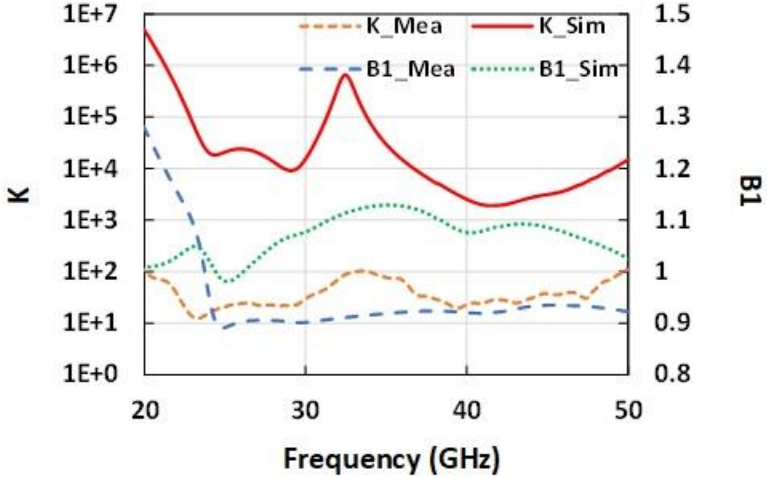
- Measurement results of standalone LNA.

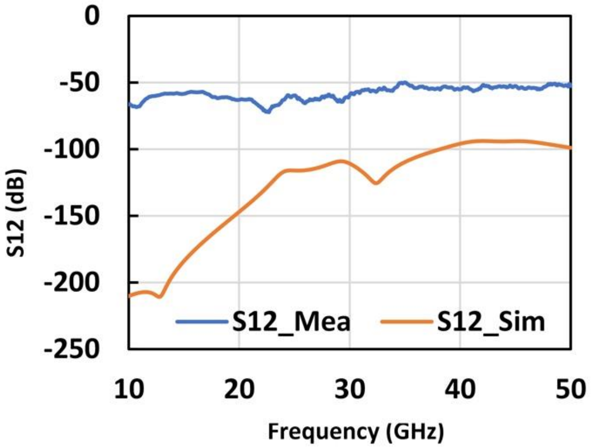


(a)

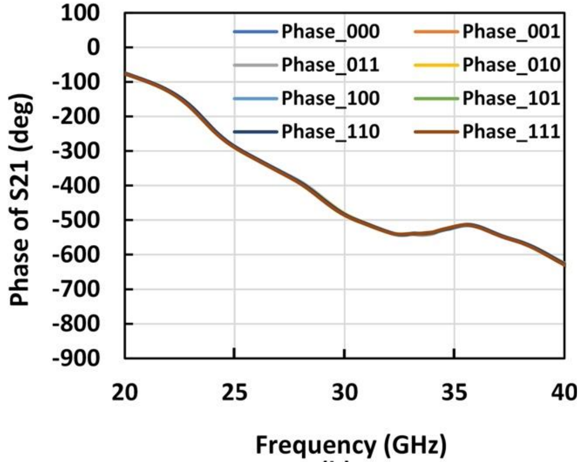


(a)





(b)



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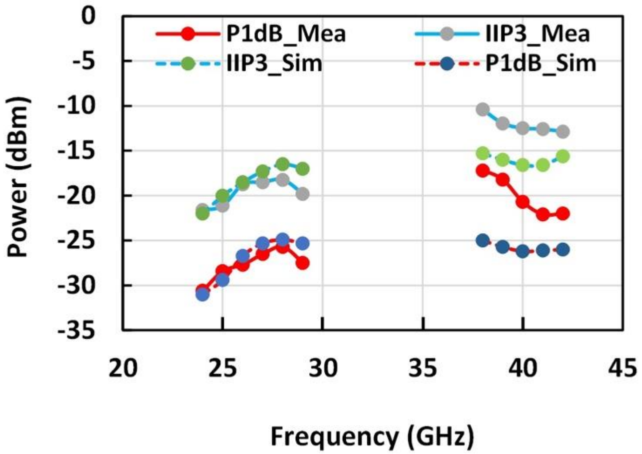

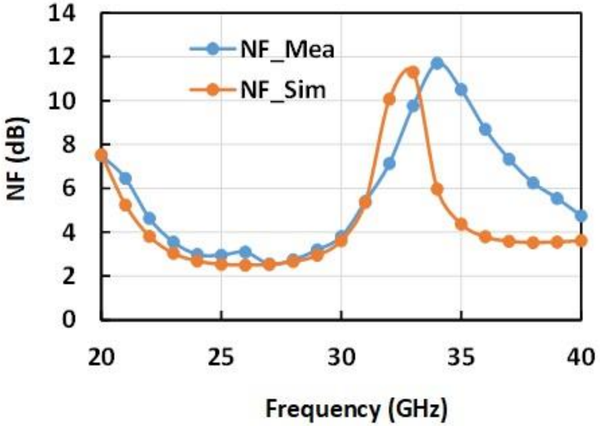

Phase Shifter Concpet

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- **Features**
- Critical part in phased-array systems
- Spatial selectivity
- Compensating free-path loss

- **Challenges in phased-array systems**
- Phase/Amplitude Accuracy
- Area
- Noise performance (stand-alone and in a system)
- Linearity
- Power consumption

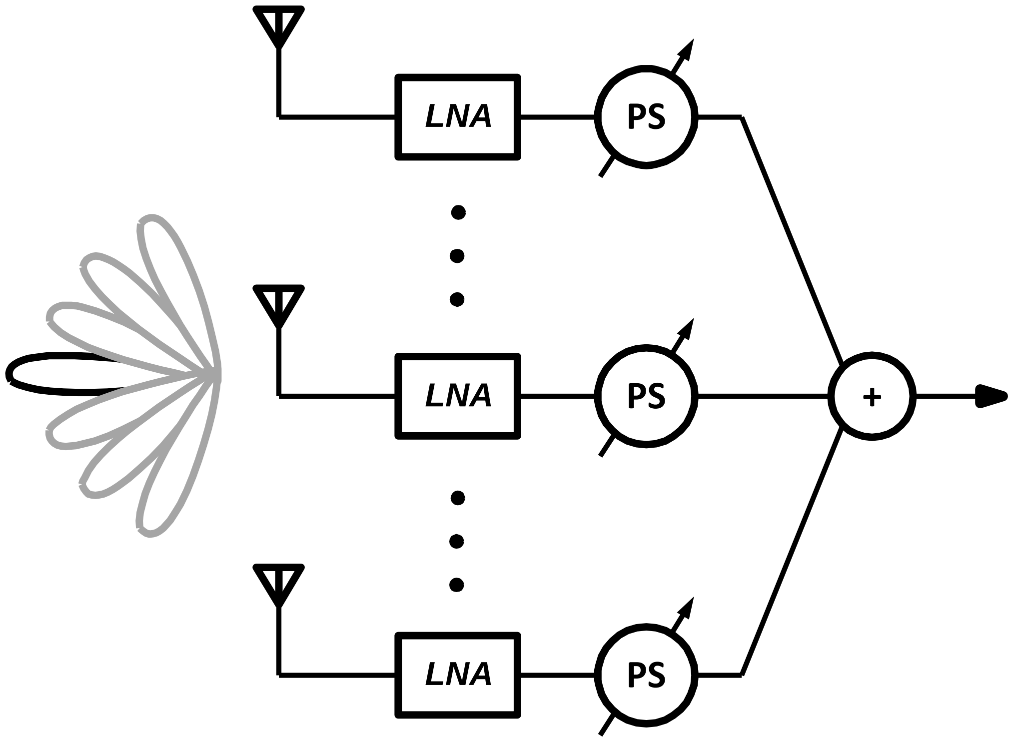

Fig. 22. The block diagram of a phased-array receiver system.

Proposed structure

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- 5-bits Active phase shifter
- Frequency Range: 24 – 36 GHz.
- Gain compensated RC poly-phase filter
- Reactance Invariant vector-modulator (VM)

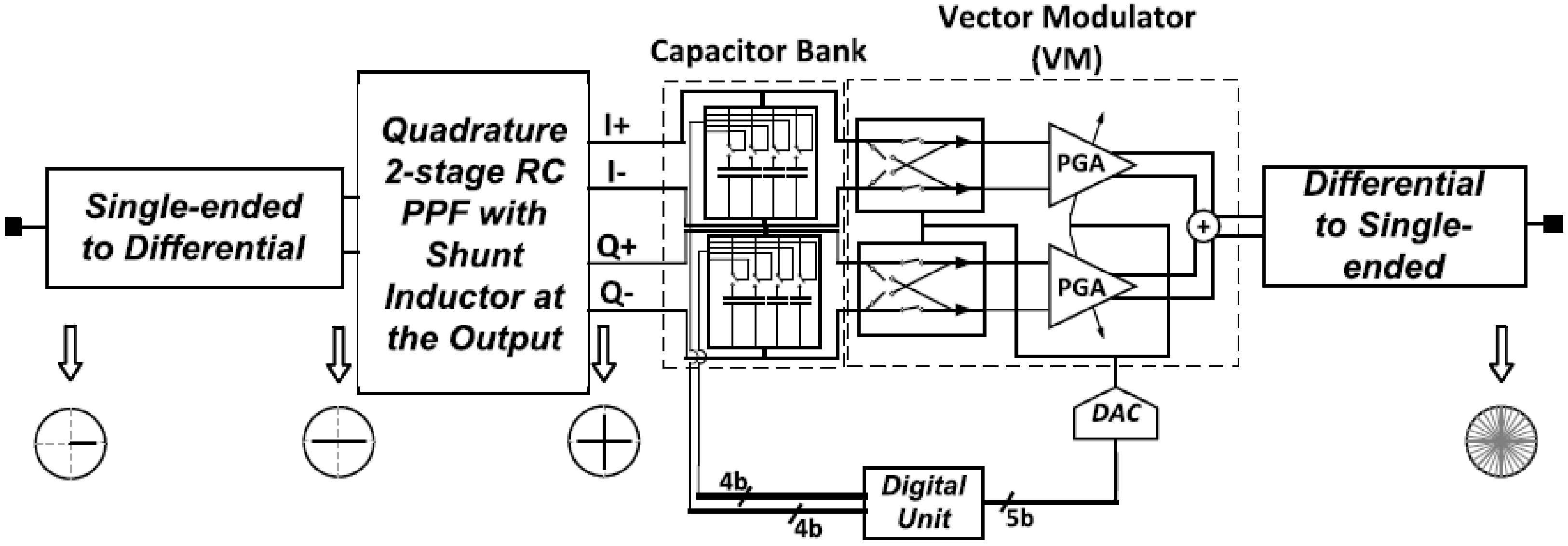
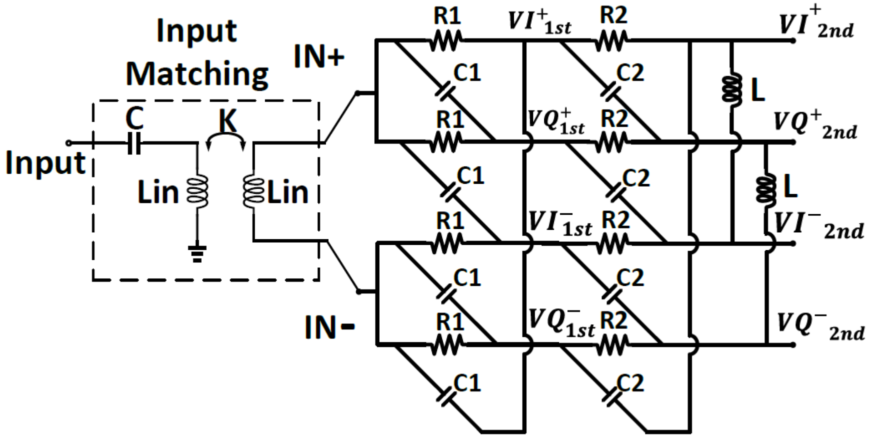


Fig. 23. The block diagram of the proposed active phase shifter.

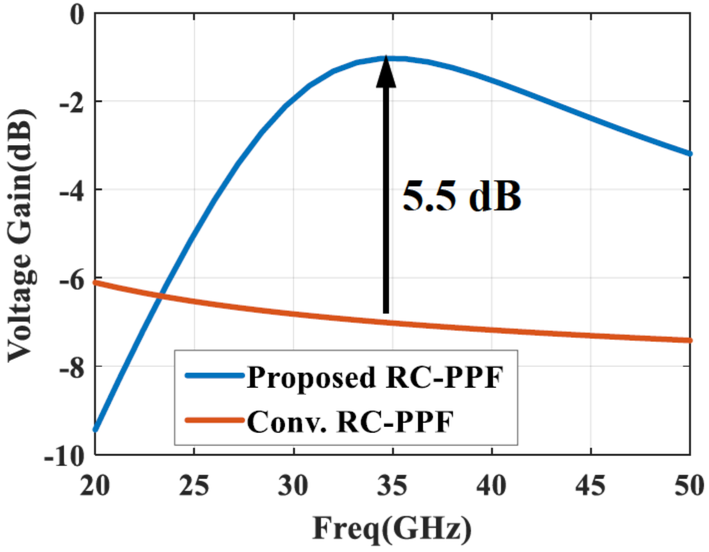
M. G. Bardeh, J. Fu, N. Naseh, J. Paramesh, and K. Entesari, “A wideband low RMS phase/gain error mm-wave phase shifter in 22-nm CMOS FDSOI,” IEEE Microwave and Wireless Technology Letters, 2023.

Proposed Structure

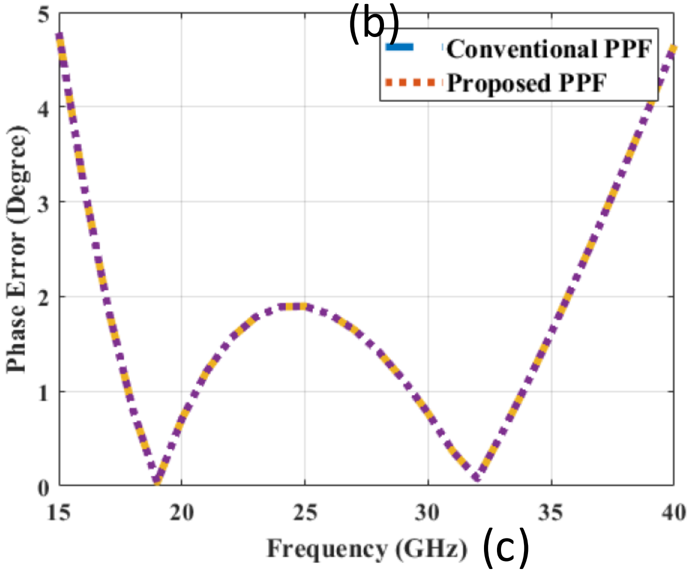
- Maintaining good features of RC poly-phase filter while compensating for loss.
- Shunt inductor
- The single-ended to differential input matching circuit



(a)



(b)



(c)

Fig. 24. (a) The schematic of the proposed gain compensated RC poly-phase filter. (b) Comparison of voltage gain response with conventional realization. (c) phase accuracy with respect to 90-degrees comparison.

Proposed Structure

- Reactance Invariant vector-modulator (VM)
- Creating 0-360° phase shifting by:
- 1) using switches S_I and S_Q, the sign
- of each I and Q signal is changed accordingly.
- 2) specific dc-current ratio between I, and Q

$$V_{out} = Z_{load}(VI \times g_{mI} + VQ \times g_{mQ})$$

$$V_{out} = Z_{load} VI \sqrt{\mu_{eff} C_{ox} W/L}(\sqrt{I_I} + j\sqrt{I_Q})$$

$M1-8$	$20\mu m/20nm$
$M9-12$	$400\mu m/600nm$
$M13-14$	$20\mu m/20nm$
R_B	$12.5\ k\Omega$
C_B	$807\ fF$
L_1	$400\ pH$
L_2	$100\ pH$
C_1	$70\ fF$
C_2	$280\ fF$
R_1	$200\ \Omega$
I_{REF}	$50\ \mu A$
C_n	$5\ fF$
M_n	$10\mu m/20nm$

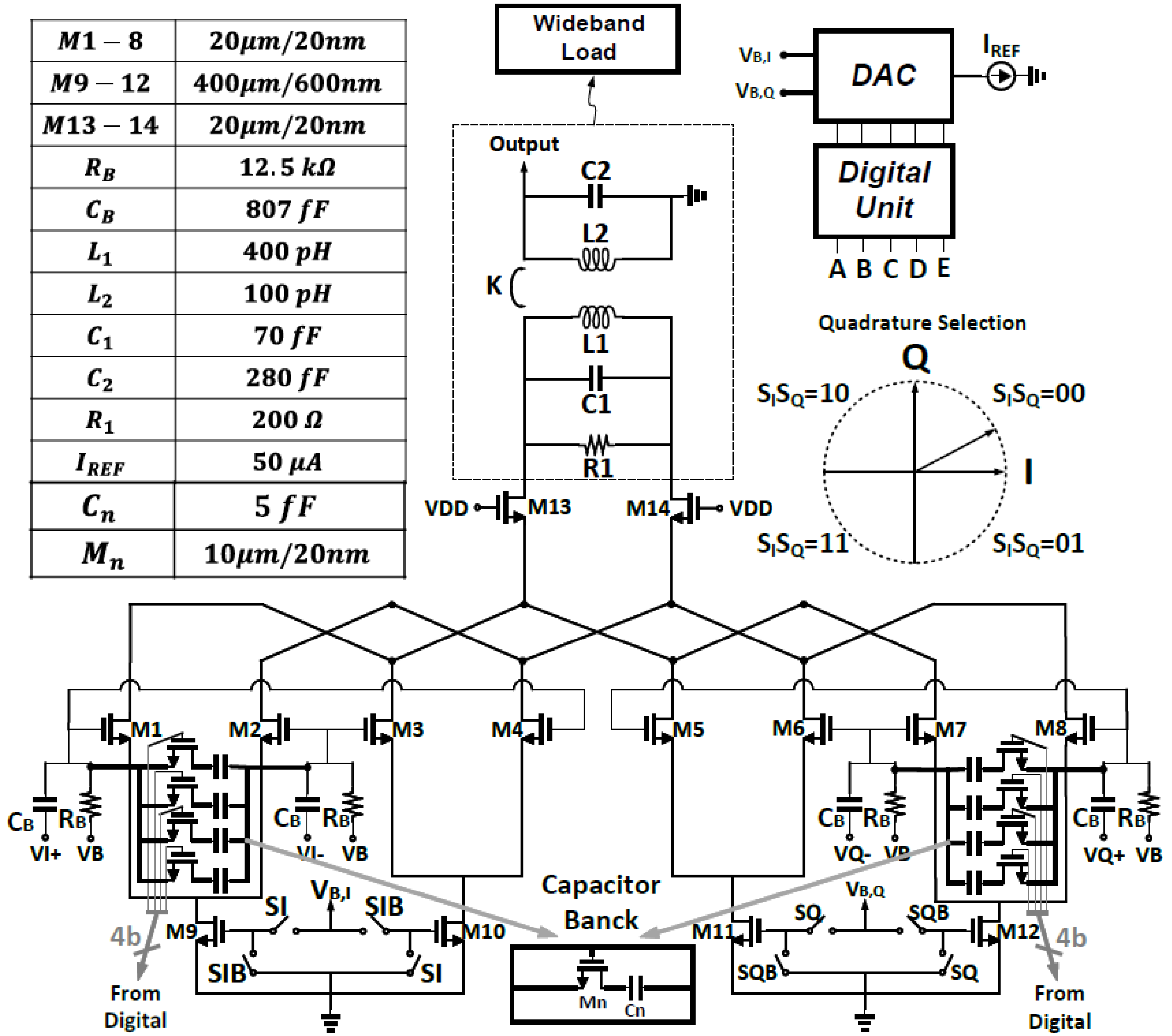


Fig. 25. The block diagram of the proposed reactance invariant vector-modulator (VM).

Fabrication and Measurement

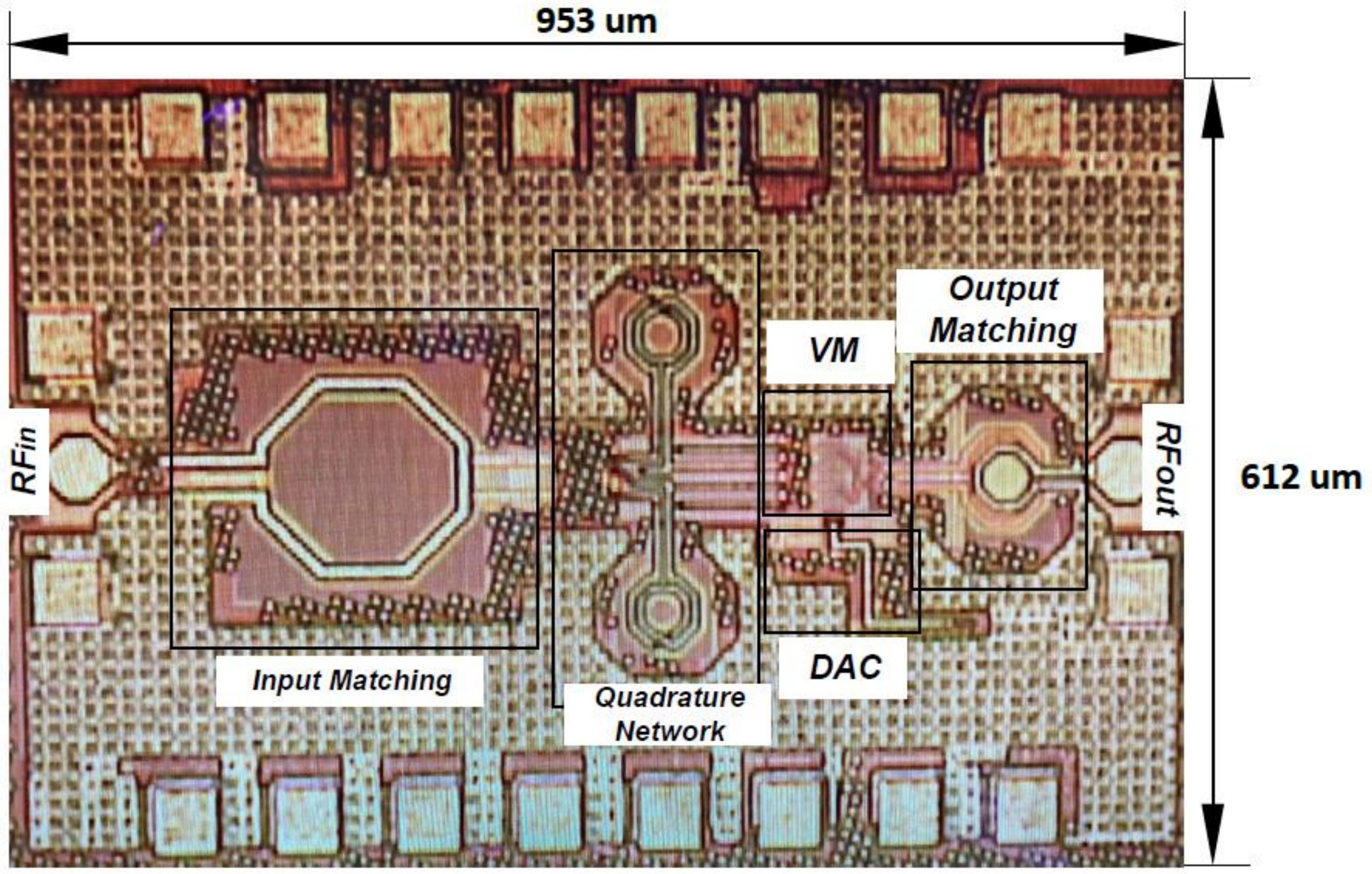


Fig. 27. Die photo of the stand-alone PS

Fabrication and Measurement

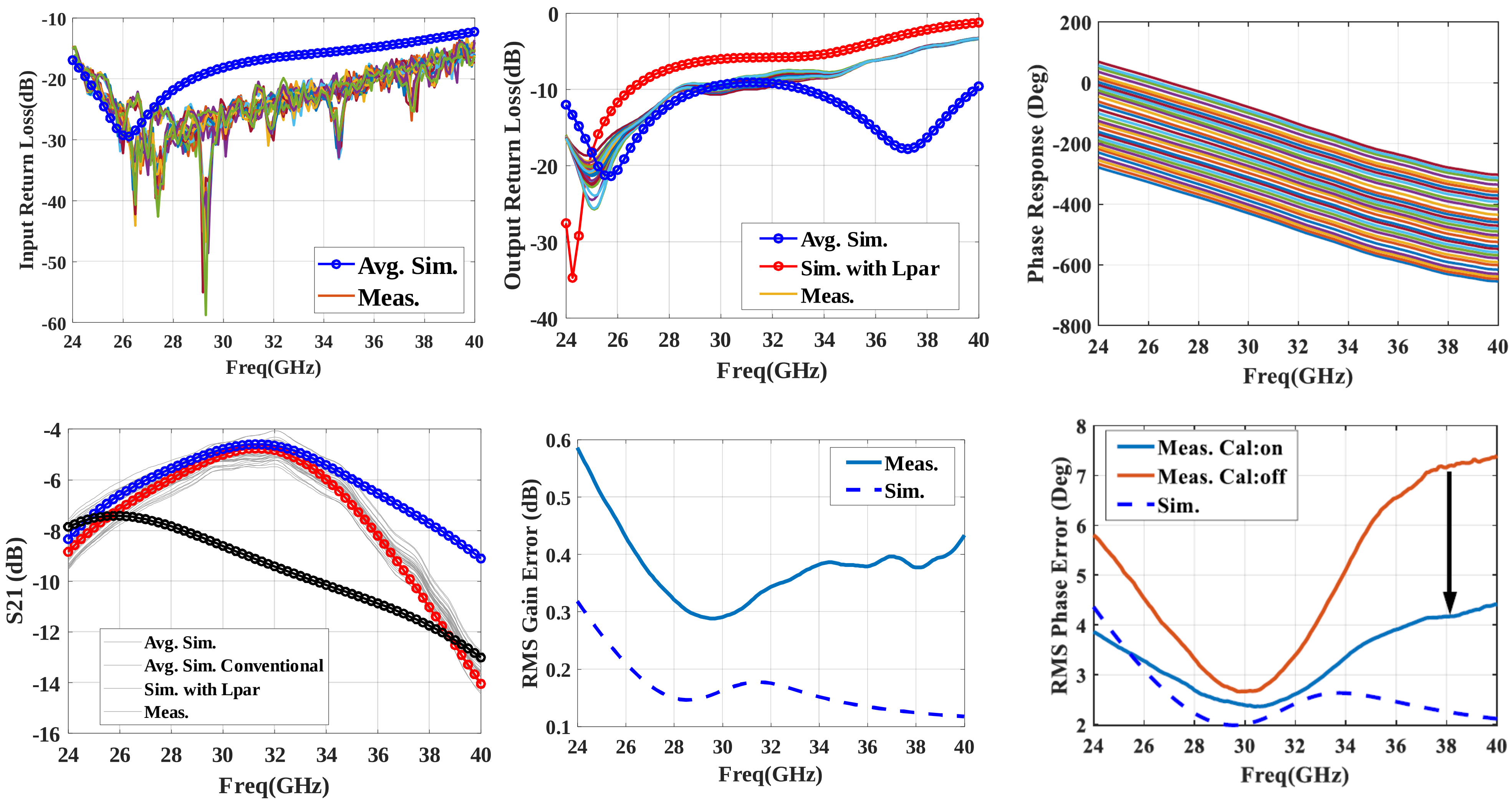


Fig. 28. Measurement and simulation results of the proposed PS.

IQ to VMs Interface

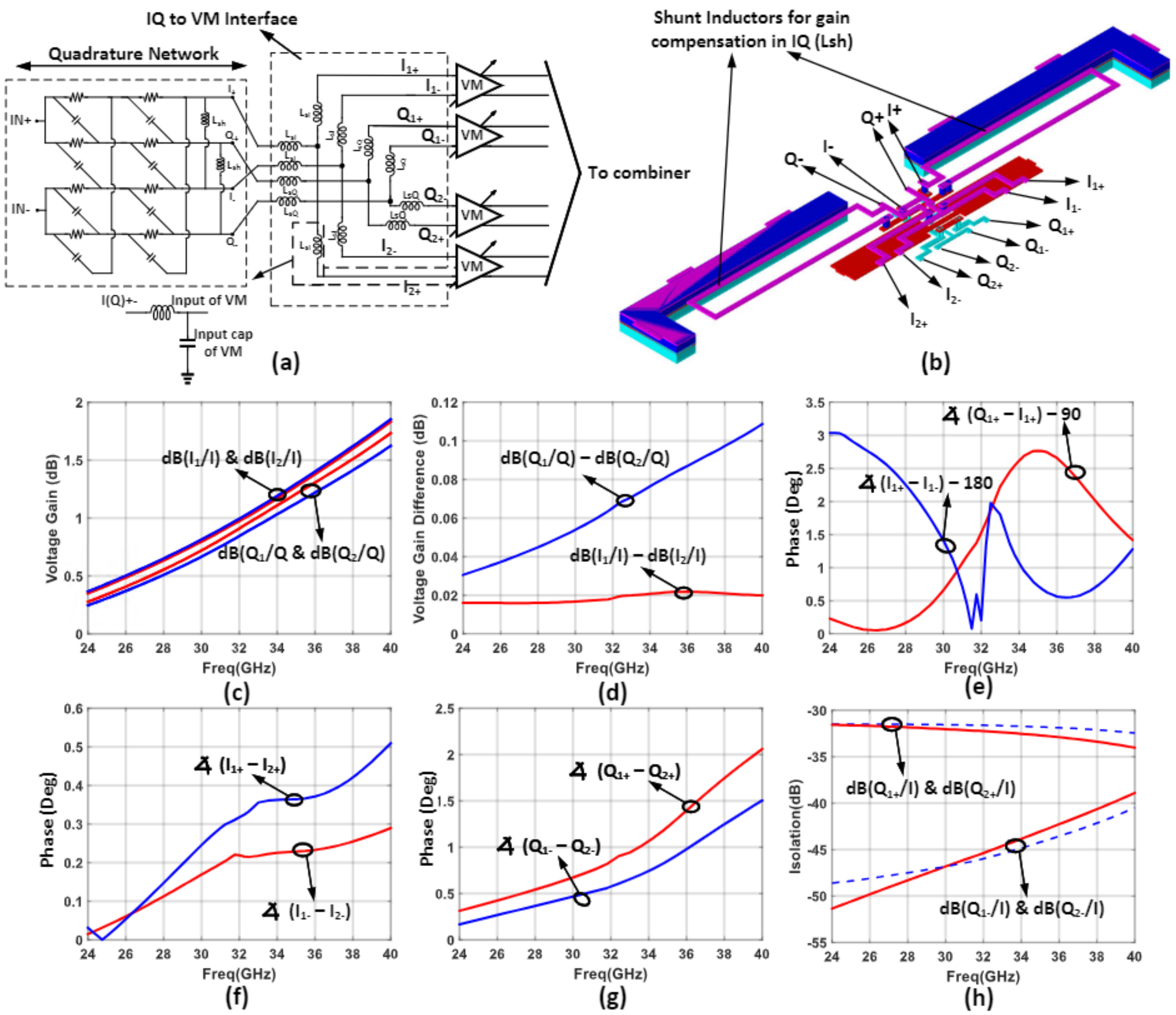


Fig. 29. Layout and simulation results for interface from IQ network to VM.

(a) The simplified schematic of the IQ, VM, and the interface.

(b) The 3D view layout of the proposed interface and the shunt inductor for gain compensation at IQ network.

(c) Voltage gain of differential I and Q paths of two VMs.

(d) Voltage gain difference of I_1(Q_1) and I_2(Q_2) paths.

(e) Phase difference of quadrature and differential signals with respect to 90 ° and 180 ° respectively.

(f) Phase imbalance of two I paths.

(g) Phase imbalance of two Q paths.

(h) Isolation between I and Q paths when I is excited at the input of interface.

Interface of VM to Combiner

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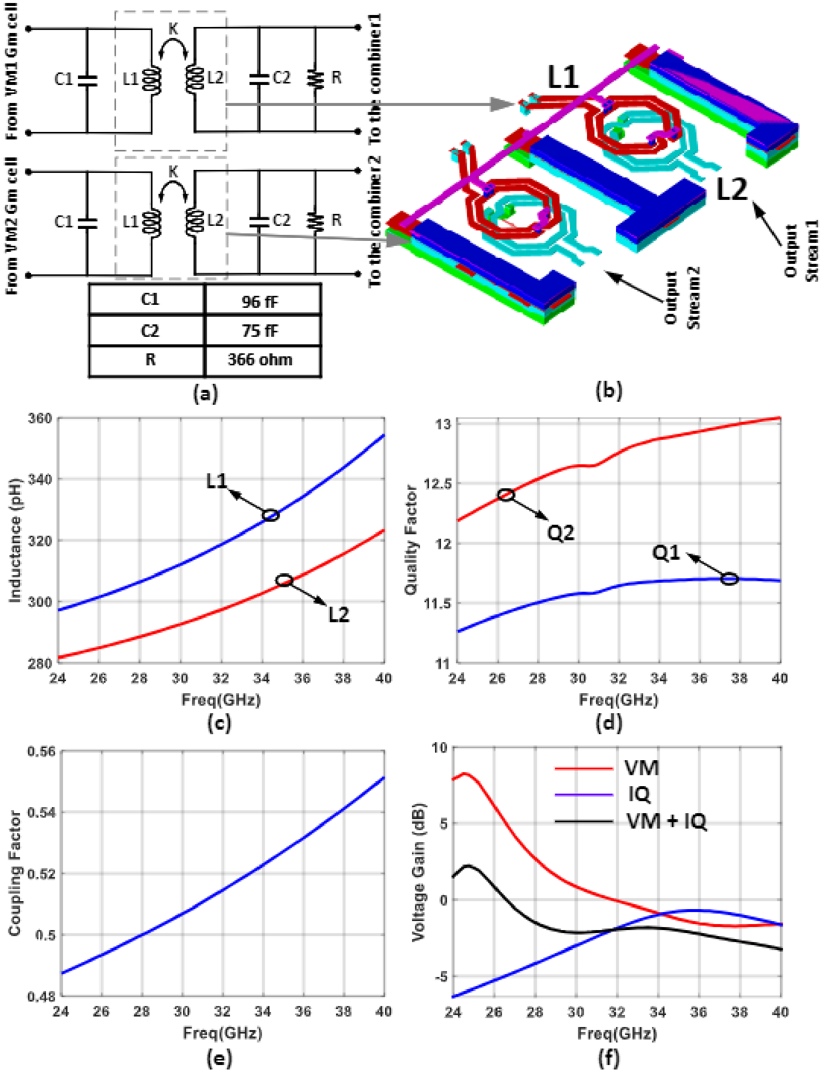

Fig. 30. Layout and simulation results of interface between the VMs and combiner.

(a) Double-tuned transformer used between output of the VM and input of combiner along with the values of capacitors and the resistor.

(b) 3D view layout of the wideband load (interface between VM G_m cell and the combiner).

(c) Inductance values of two inductors L_1, and L_2.

(d) Quality factor of inductors L_1, and L_2.

(e) Coupling factor of the transformer.

(f) Voltage gain of VM, IQ network, and VM+IQ.

Combiner

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- Wideband distributed active combiner
- 4-channels combining coherently
- Transmission line between M1 and M2 for matching
- Differential to single-ended matching network

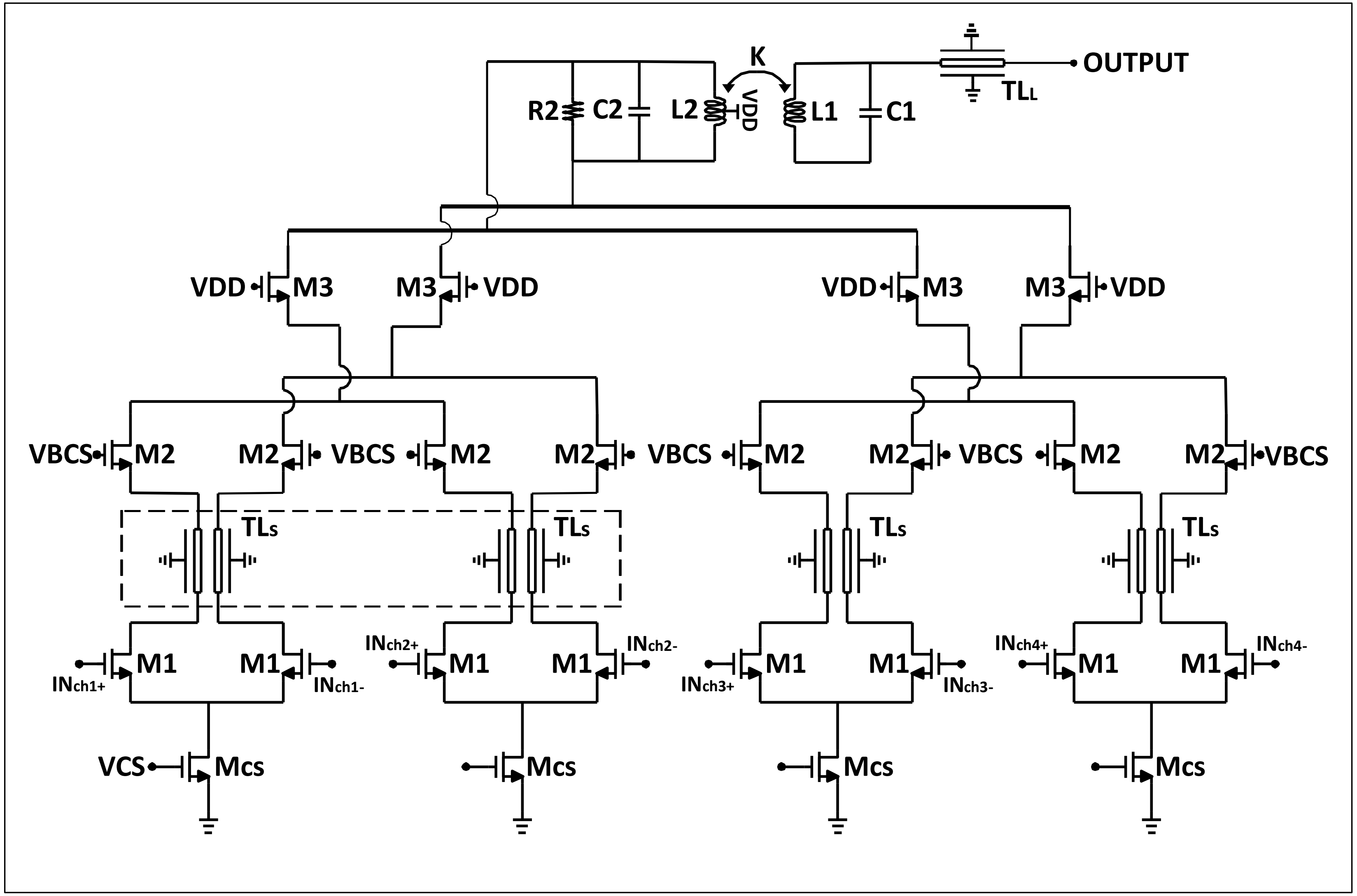
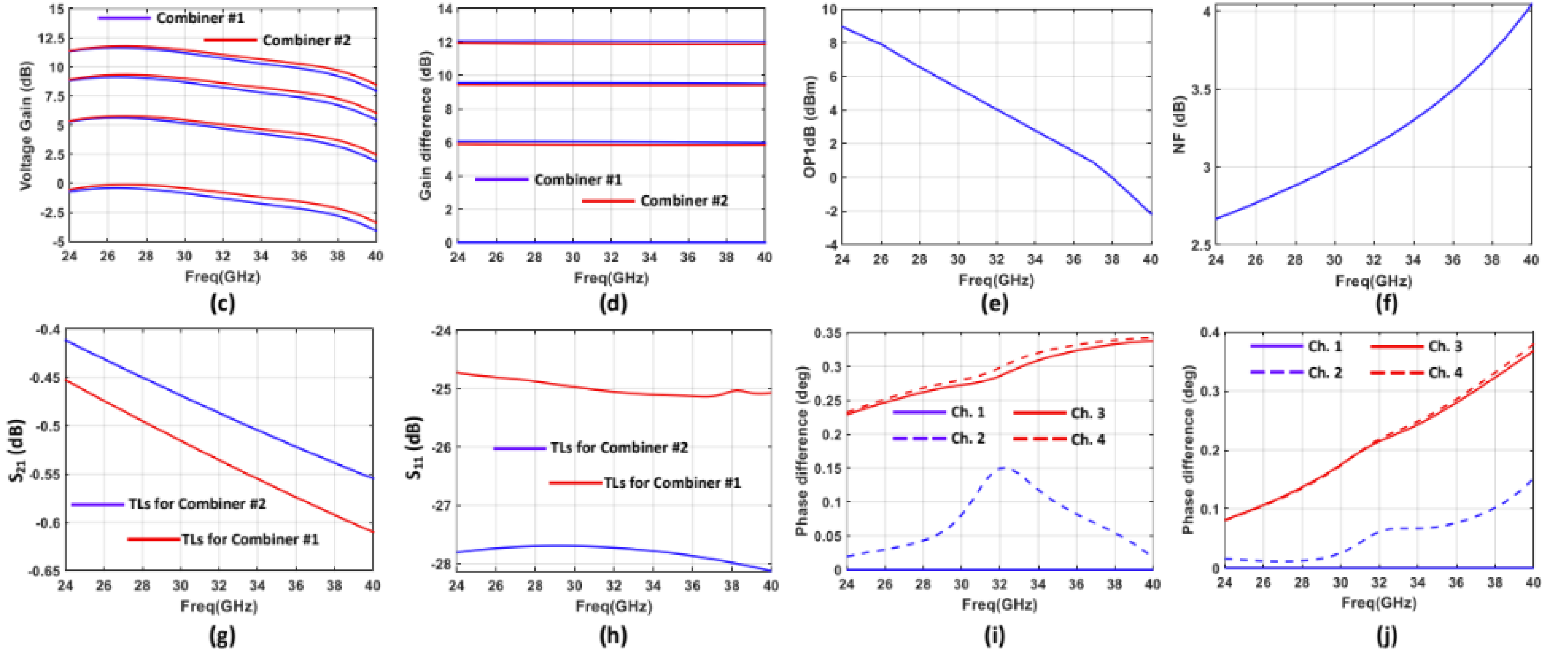


Fig. 31. The schematic of the proposed wideband distributed active combiner.

Combiner



Schematic, layout, and simulation results of the proposed distributed active combiner. (a) The schematic of the proposed combiner and its corresponding table. (b) The 3D view layout and cross section view of the proposed TLs for combiner1 and combiner2 (The layout is shown for half of the combiner routings. Also, in this view QB is removed and is shown in cross-section view). (c) Voltage gain of the combiner when signal is applied to one, two, three, or four channels. (d) Voltage gain difference for different voltage gains. (e) Output P1dB, (f) NF of the combiner vs frequency. (g) Insertion loss of the proposed TLs for combiner1 and combiner2. (h) Input return loss of TLs for combiner1 and combiner2 when Z0 = 38 Ω for a terminated 38 Ω load. (i) Phase difference at the output of combiner1 when each time only one channel is ON and others are OFF based on channel one of combiner1. (j) Phase difference at the output of combiners2 when each time only one channel is ON, and others are OFF based on channel one of combiner2.

Combiner

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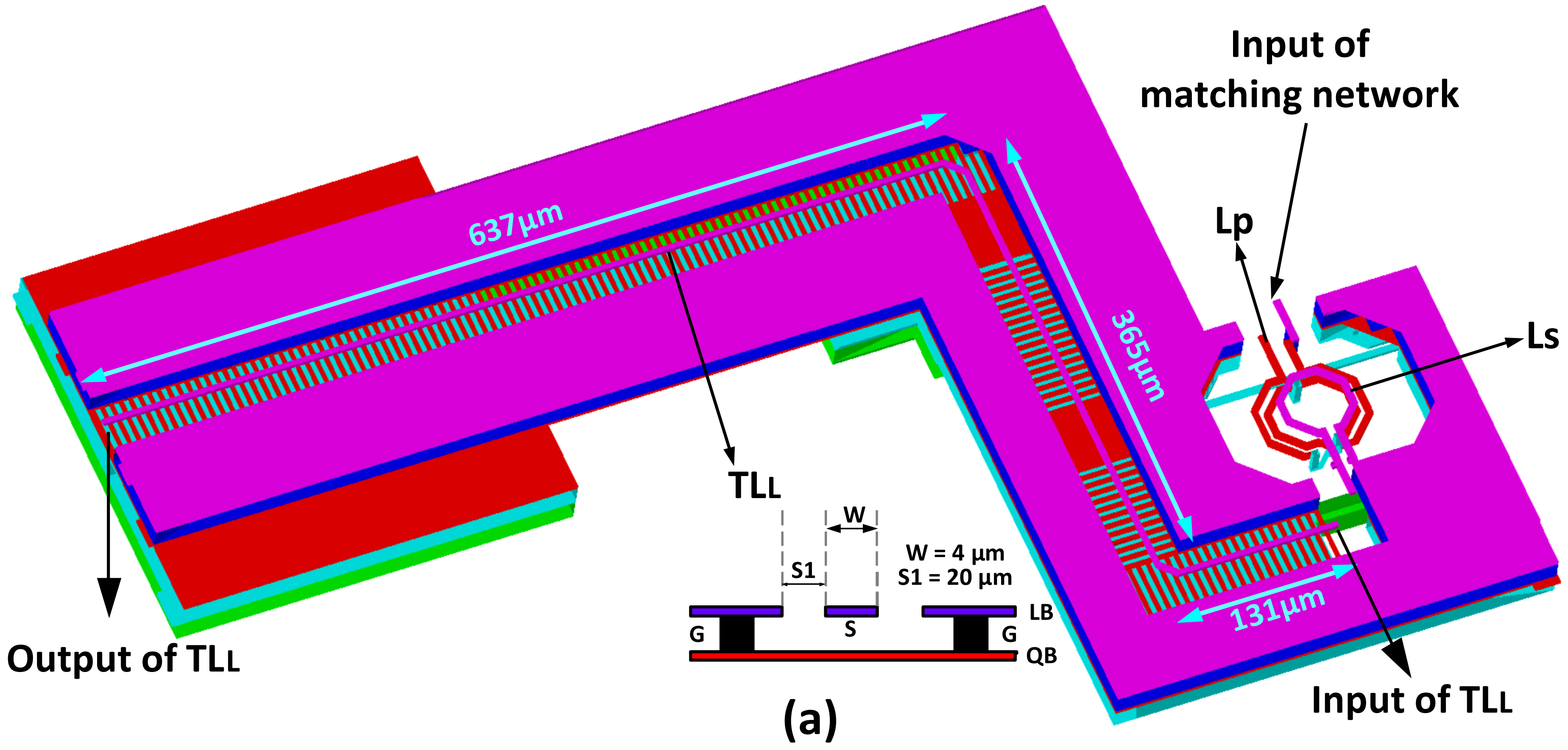

Fig. 33. The schematic of the proposed output transmission line along with output matching network of the combiner.

Combiner

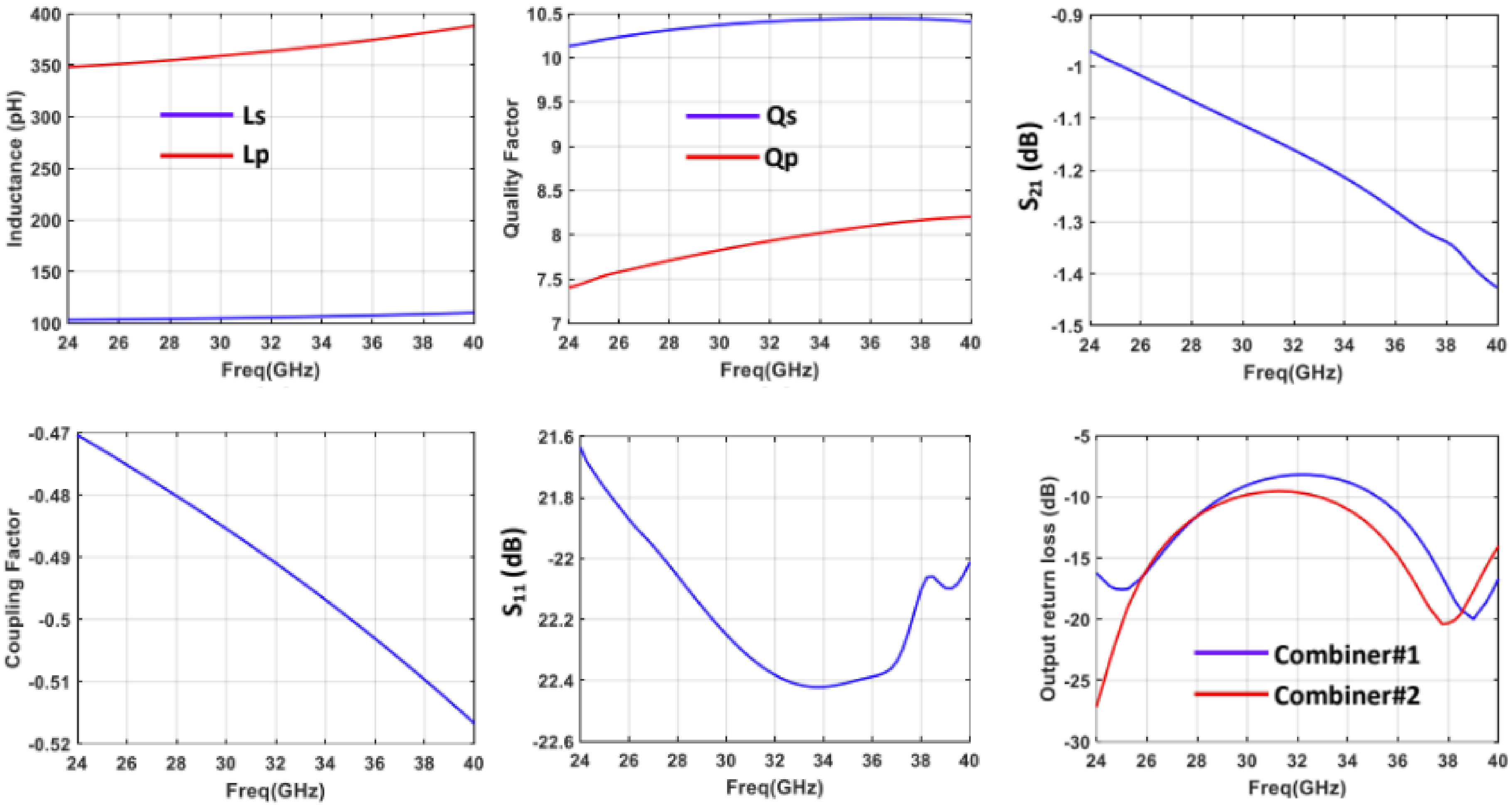

Fig. 34. Simulation results of the output matching network of the combiner.

Outline

Fabrication and Measurement

- **Measurements**
- Single-channel characterization
- Beam Steering Measurement
- Null Tuning
- Measurement with Modulated Signals

- **Measurement setup**
- Rohde & Schwarz ZVA67 network analyzer
- Rohde & Schwarz FSV40 spectrum analyzer
- Pasternack PE85N1008 noise source
- ETS-Lindgren 3116C double-ridged waveguide horn antenna
- Form-Factor InfinityQuad multi-contact RF probes
- Form-Factor probe station.

Fabrication and Measurement

- 4 Low noise amplifiers (LNA)
  - The majority of this block is done by Jierui Fu.
- 4 Phase shifters (PS)
  - 4 Quadrature network (QN)
  - 8 Vector-modulator (VM)
- 2 Combiners
- Global Foundry 22 nm Fully Depleted Silicon On Insulator (FDSOI) CMOS

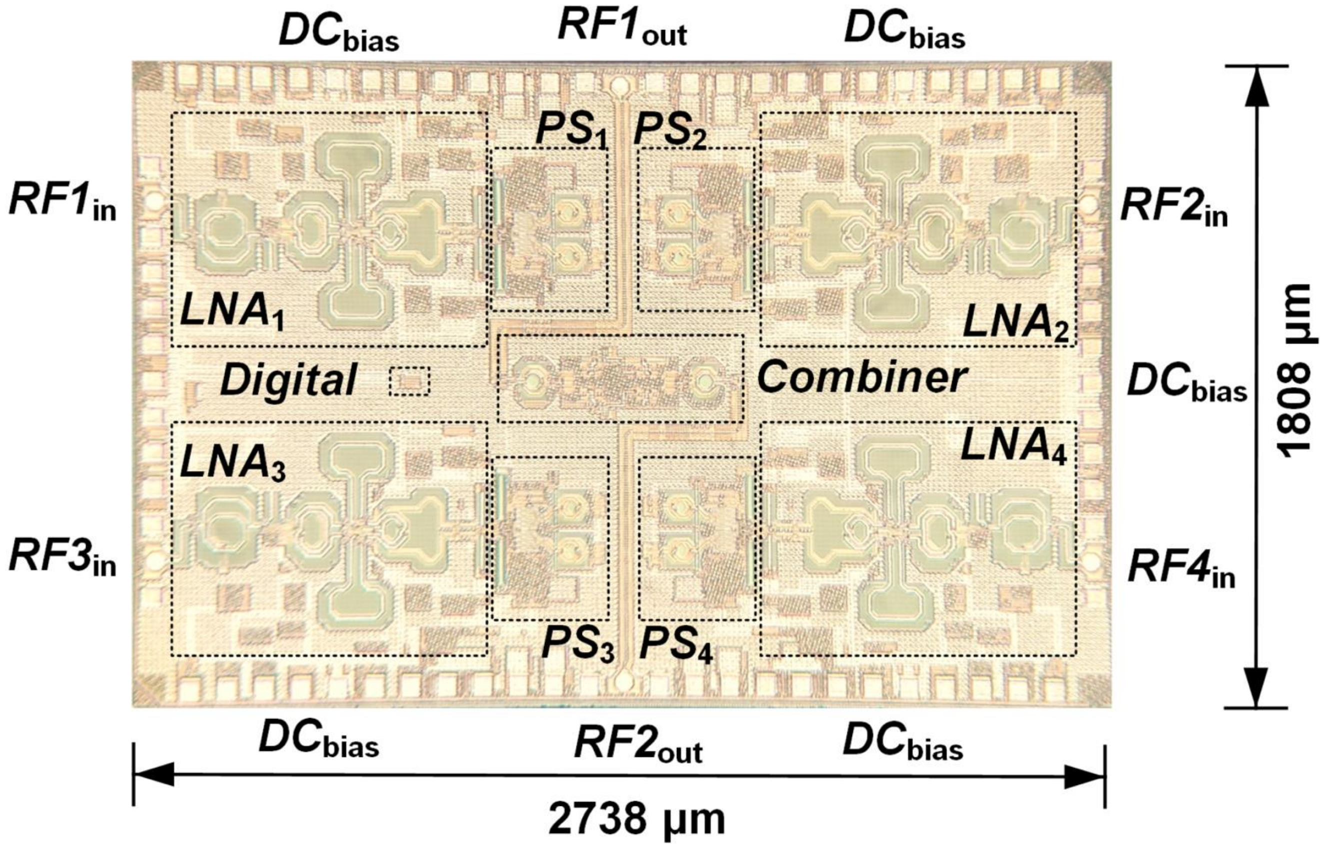

Fig. 35. Die photo of the concurrent dual-band dual-beam phased-array receiver.

Fabrication and Measurement

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- **Single-channel characterization setup**

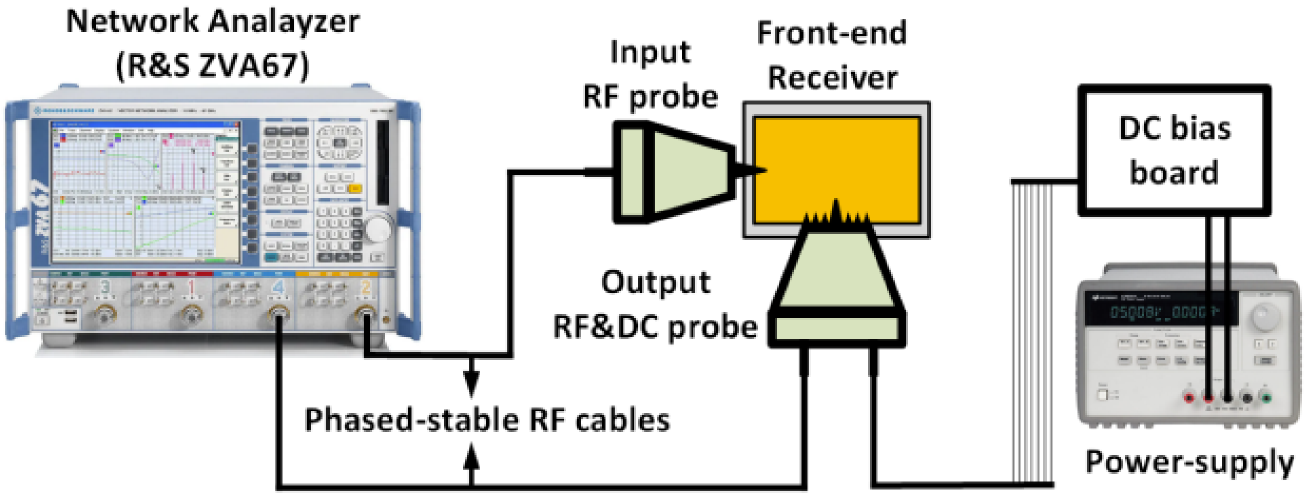

Fig. 34. Single-channel characterization setup.

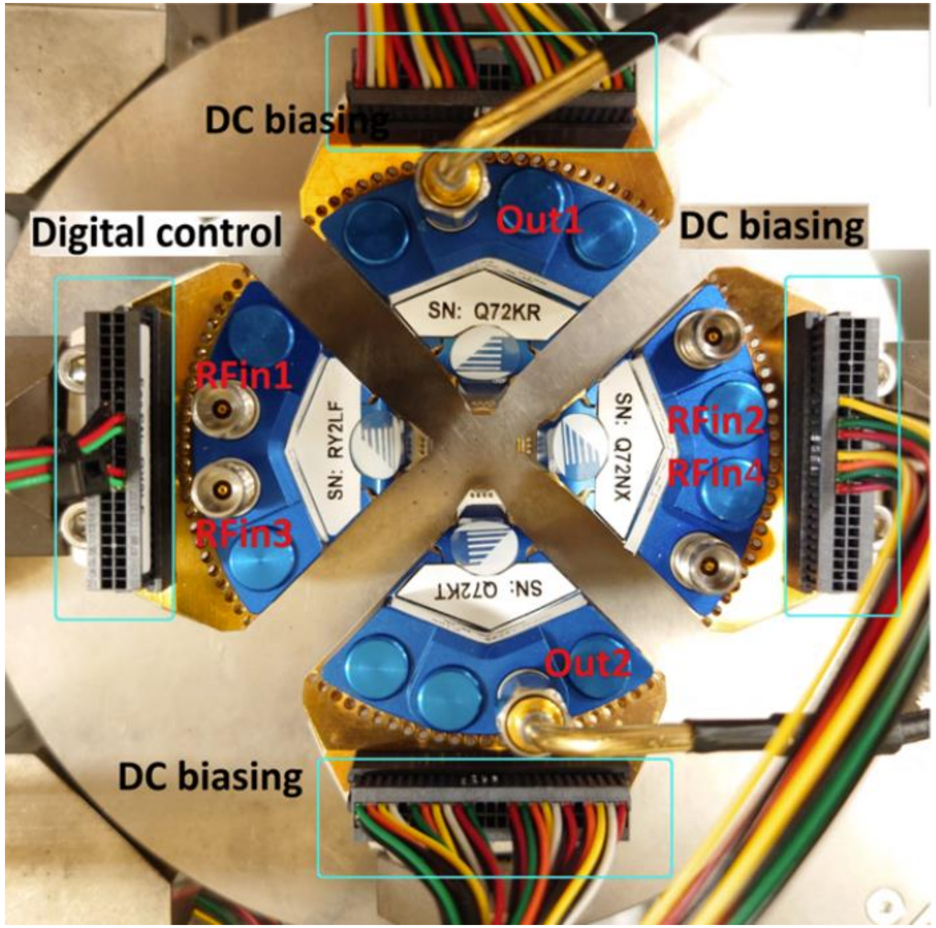

Fig. 36. The top-view of the RF and DC probes.

Fabrication and Measurement

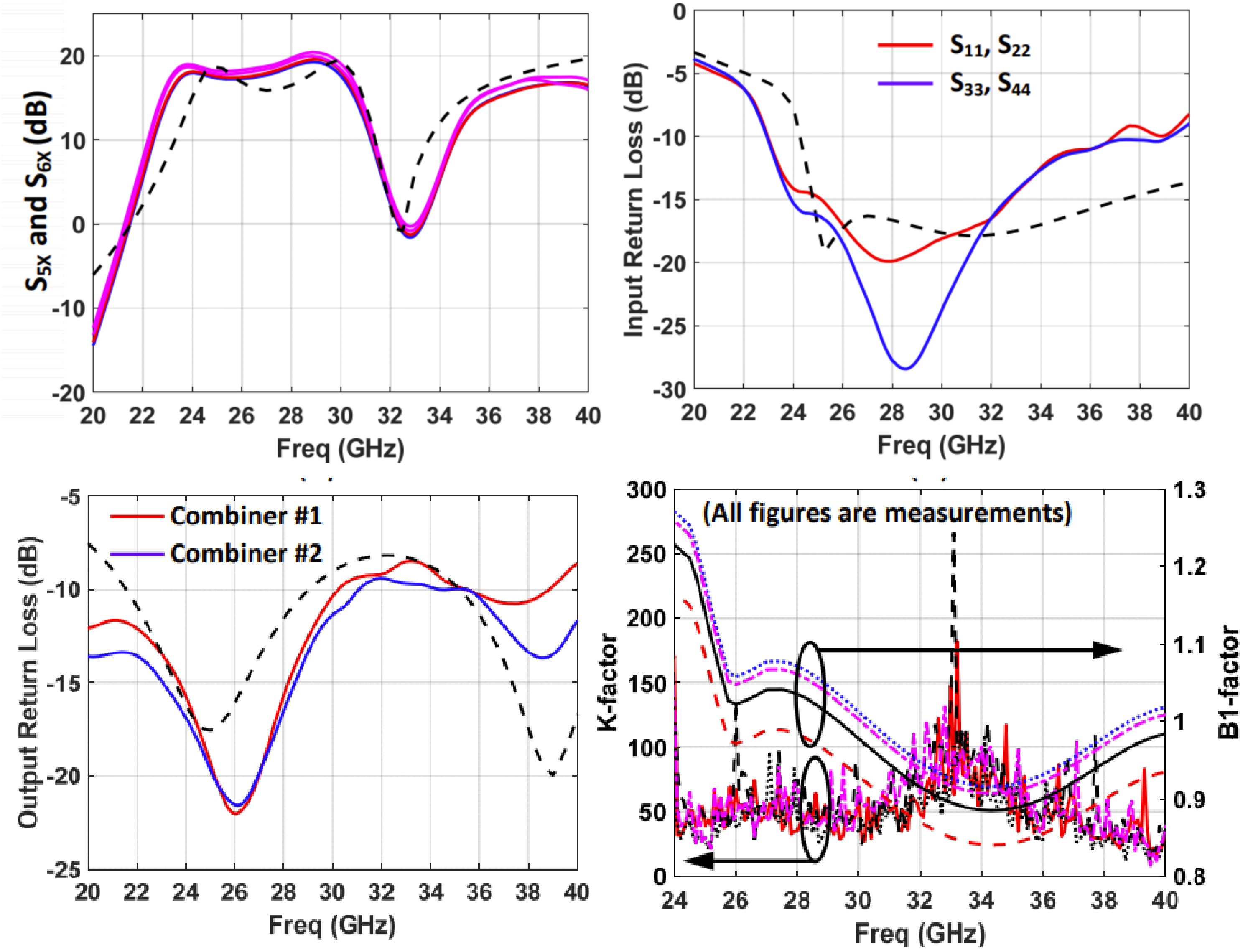

Fig. 37. Measurement of single-channel characterization results.

Fabrication and Measurement

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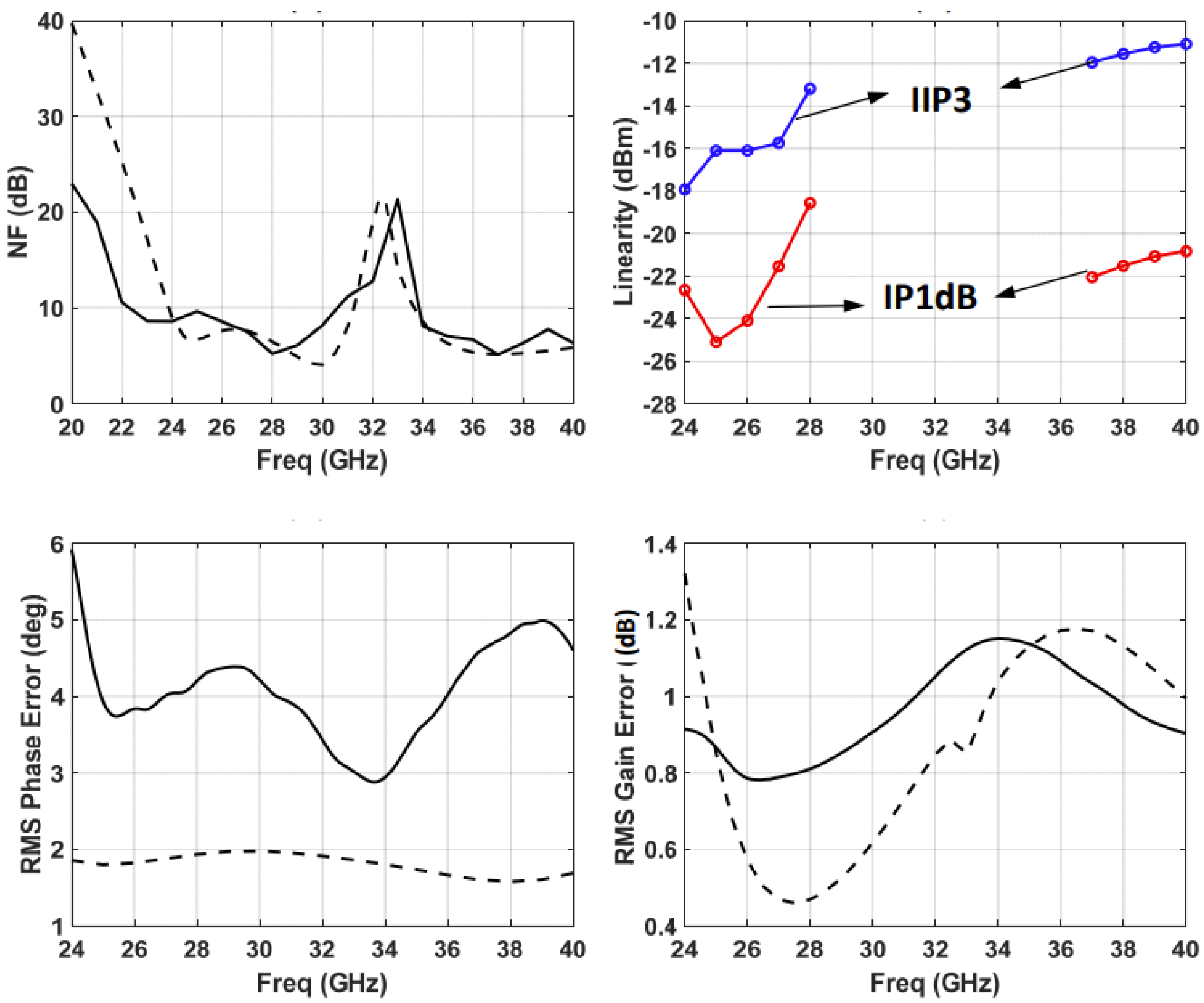


Fig. 38. Measurement of single-channel characterization results.

Fabrication and Measurement

- **Over-the-air Measurement Setup**

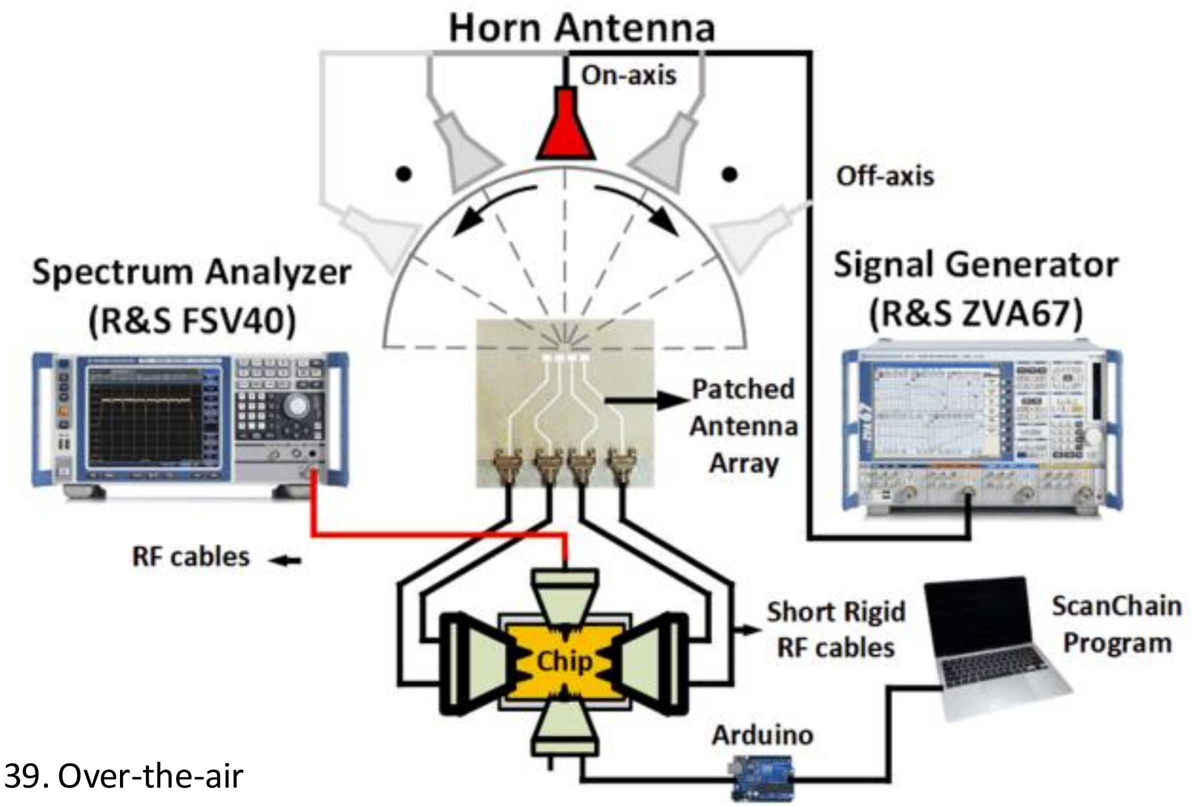
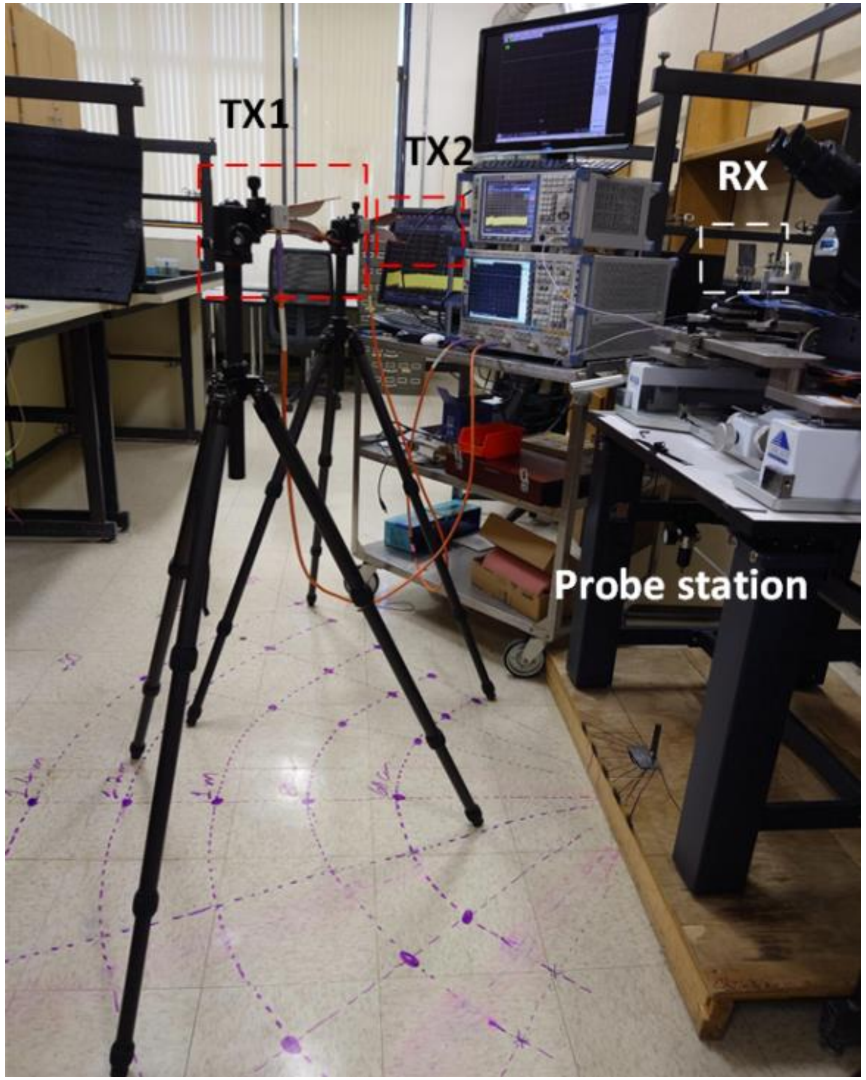
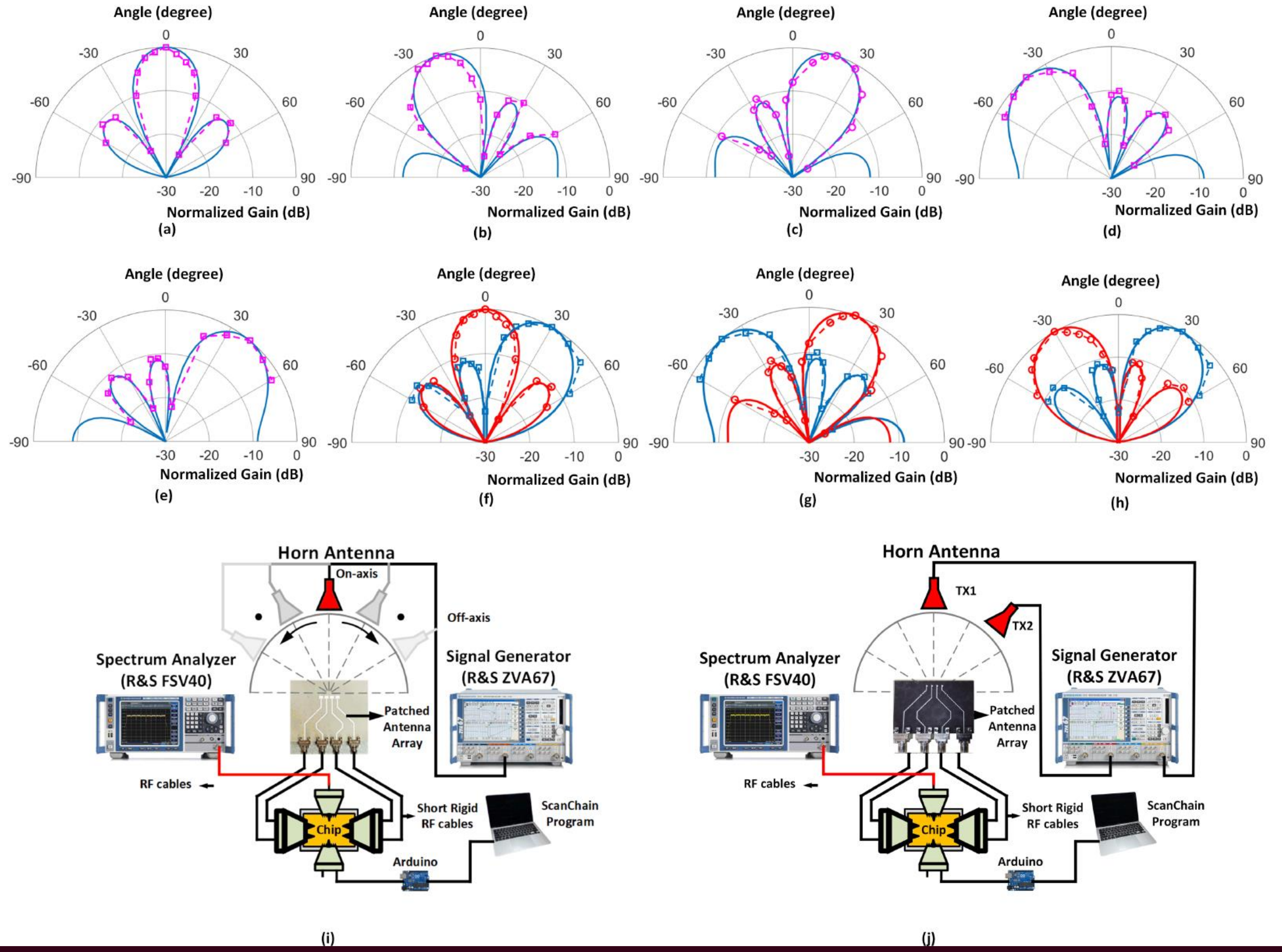


Fig. 39. Over-the-air measurement setup.



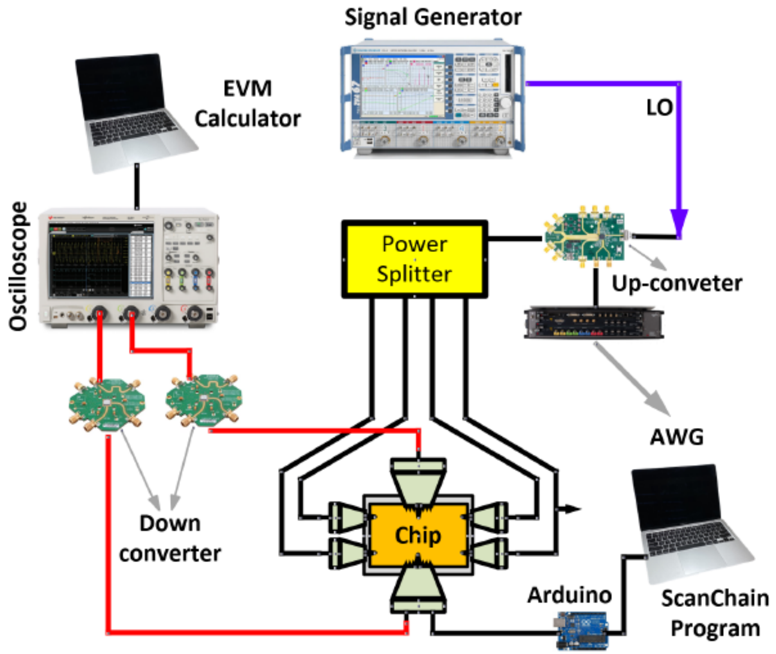
Fabrication and Measurement

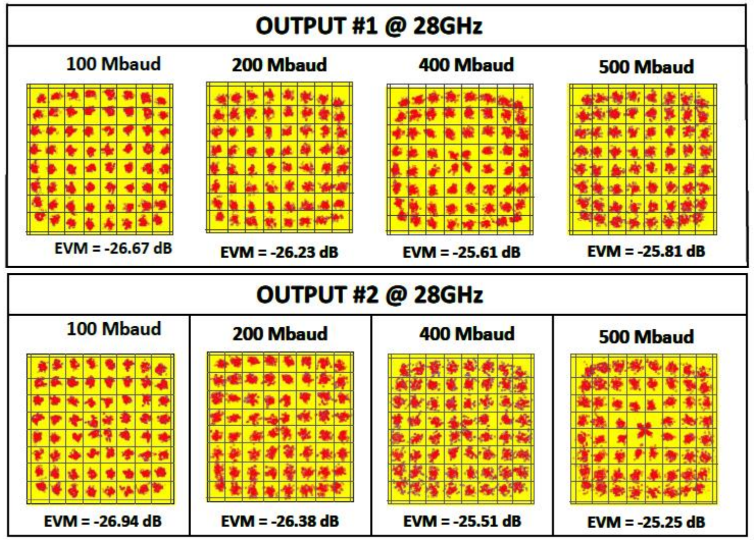
- Measurement results of Phased array, beam steering.



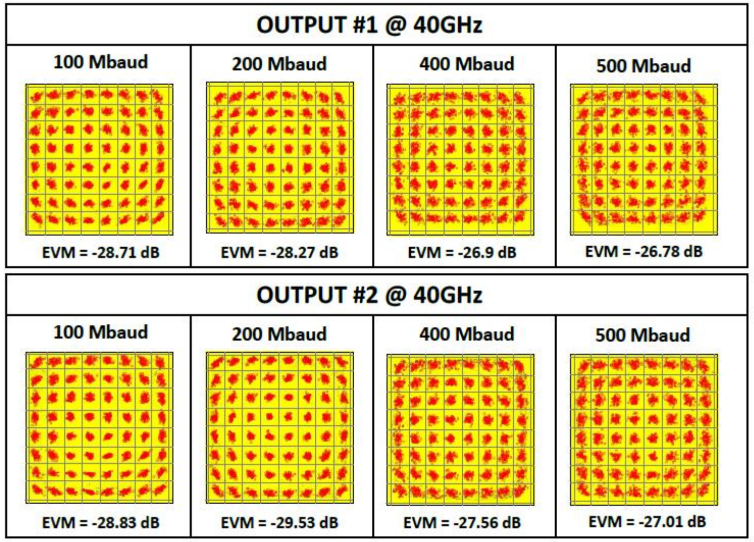
Fabrication and Measurement

- Measurement results of Phased array, EVM with cables.





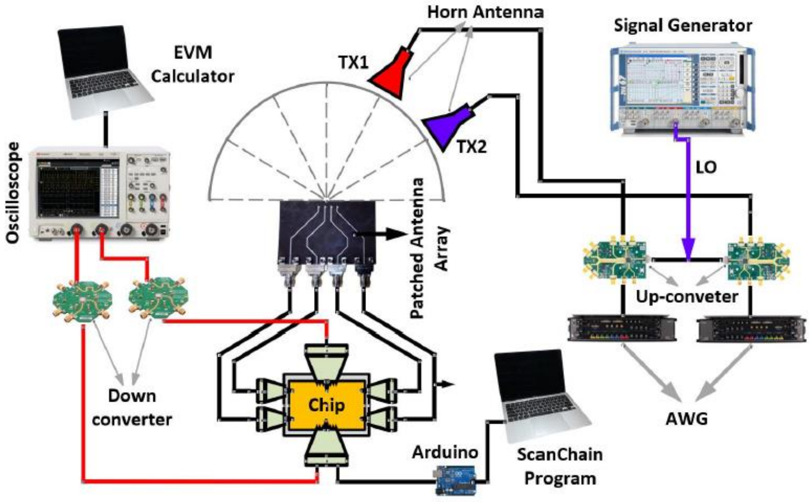
(a)

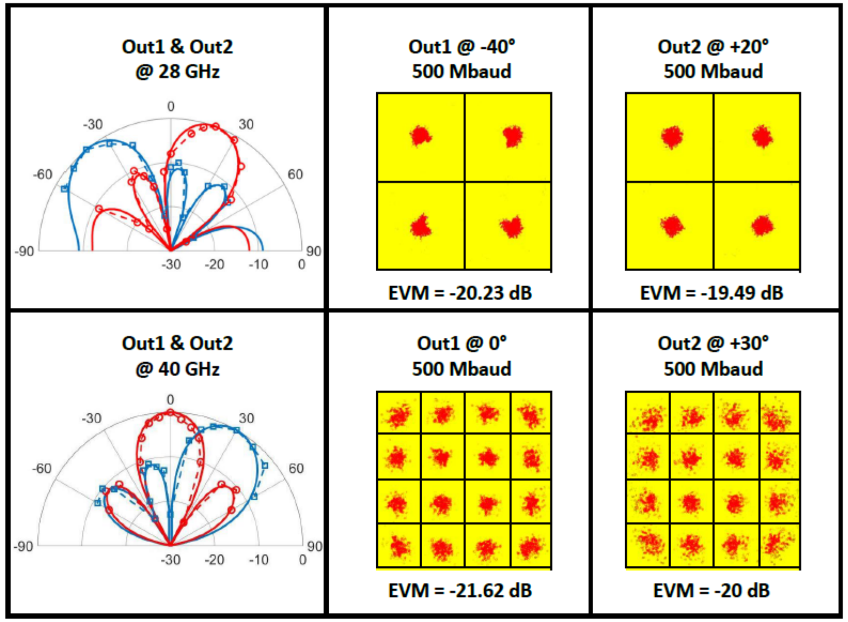


(b)

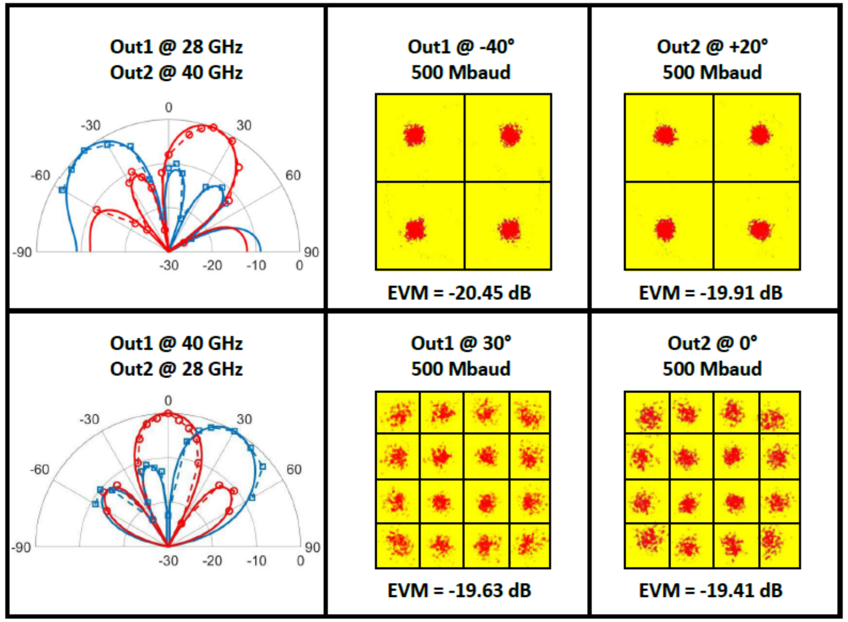
Fabrication and Measurement

- Measurement results of Phased array, EVM over-the-air.





(c)



(d)

Outline

I. Introduction
II. System Architecture
    A. Array literature review
III. Proposed Array Architecture
    A. Circuit Implementation
        1. Low Noise Amplifier (LNA)
        2. Phase Shifter (PS)
        3. IQ to VMs Interface
        4. Interface of VM to Combiner
        5. Wideband Distributed Active Combiner
    B. Fabrication and Measurement
IV. Conclusion

Conclusion

- To enable multi-band MIMO communication in multi-antennas systems:
- Fully integrated mm-Wave concurrent dual-band dual-beam phased array receiver front-end
- The unique phase shifter design which enables simultaneous RF-beamforming for two independent output streams,
- The front-end architecture utilizes a dual-band frequency response with mid-band rejection which achieves optimized bandwidth for lower and higher bands and canceling spectral interfere right after LNA. In terms of circuit-level.
- A wideband mm-Wave receiver using a wideband/reconfigurable LNA and on-chip image-rejection circuitry with high image-rejection-ratio (IRR) is presented.
- Proposed a novel 3-winding transformer load which allow both LNA and receiver to work in two wideband and reconfigurable modes.
- Enabling out-of-band interfere signal rejection
- The complete analysis of the proposed 3-winding transformer is presented.

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Thanks!